Welcome to Bite-Sized HTML5 Game Development!

Bite-Sized courses are a fresh take on our most popular content. Presented in concise, fast-paced, 3-minute lessons, they're designed to give you a quick overview of the most popular areas of programming in a short amount of time.

Ready to get started? Let's check out the first lesson!

Prefer a course that takes a more comprehensive approach? Try Create a Road Crossing Game with Phaser 3.

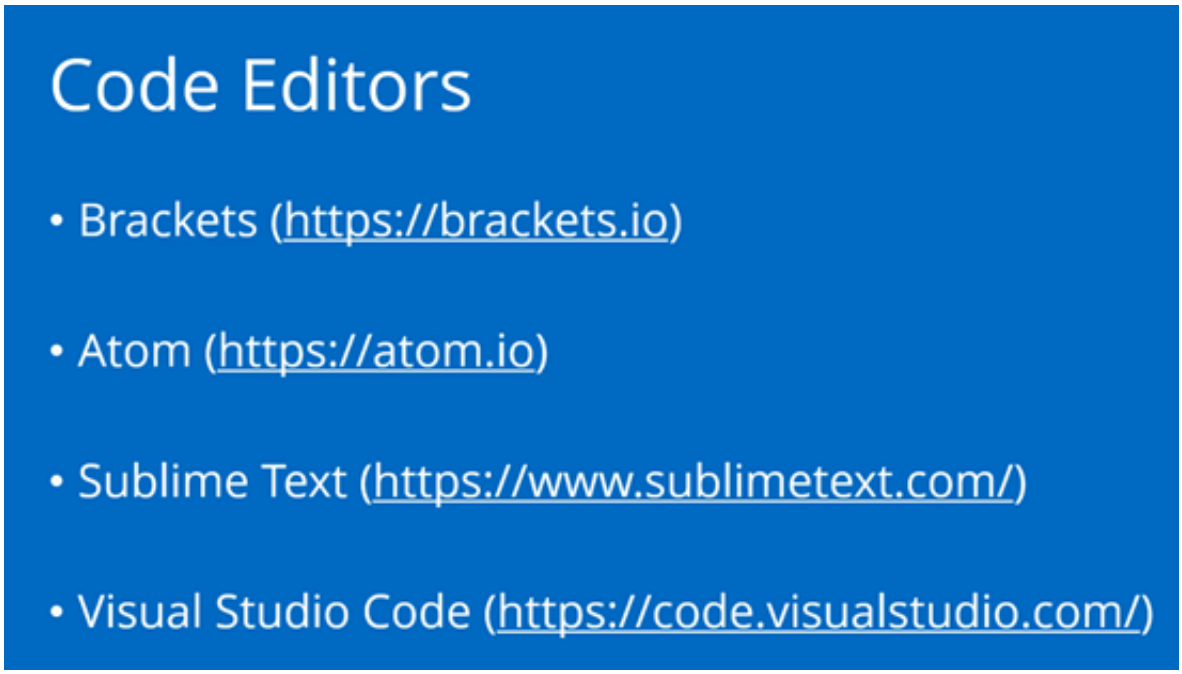
Learning Goals

- Basic Phaser development environment.
- Setting up a web server.

In order for you to start making games with Phaser you will need to have a **web browser** (Google Chrome will be used for this course), **code editor**, and a **web server**.

There are many different code editors out there that are free to use, and it doesn't matter which one you choose.

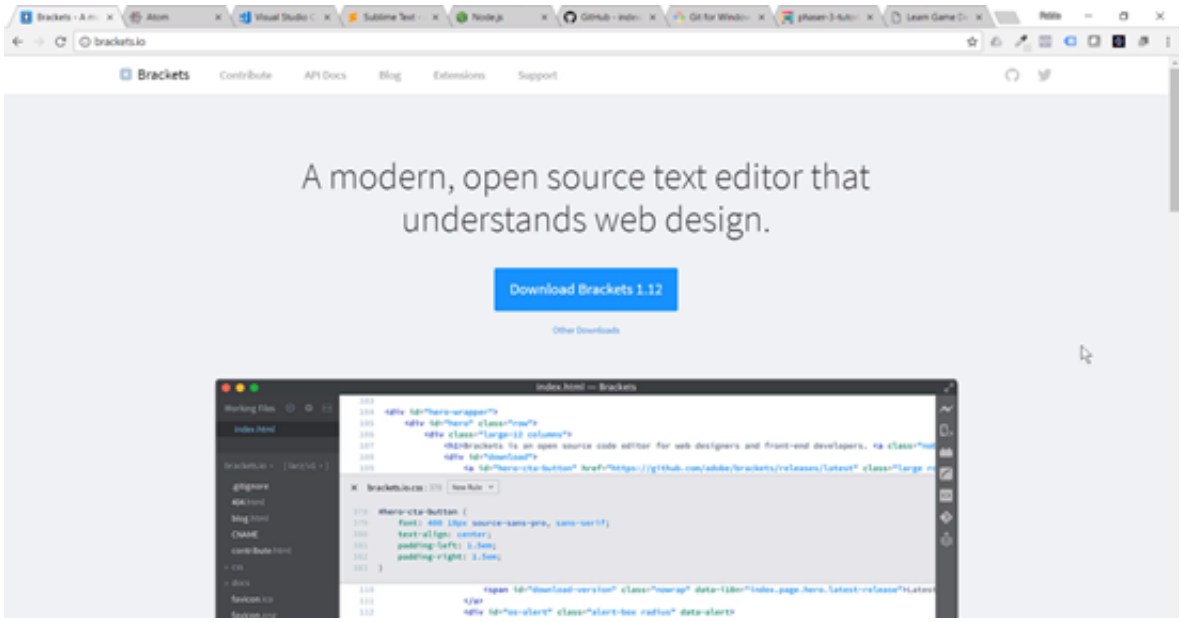
- A list of code editors that are free
    - Brackets(https://brackets.io)
    - Atom(https://atom.io)
    - Sublime text(https://www.sublimetext.com)
    - Visual Studio Code(https://code.visualstudio.com/)



A web browser is a program that will receive requests from the browser, and we will send, it will serve files as a response.

A browser doesn't allow you to load files from the file protocol due to security reasons. The web would be a very dangerous place if random websites could get access to the files on your computer. This is why you cannot just go and open a Phaser game by double clicking on the index.html file.

You have to load the game through a **web server**. The simplest way to get a web server up and running is to install the Brackets code editor.



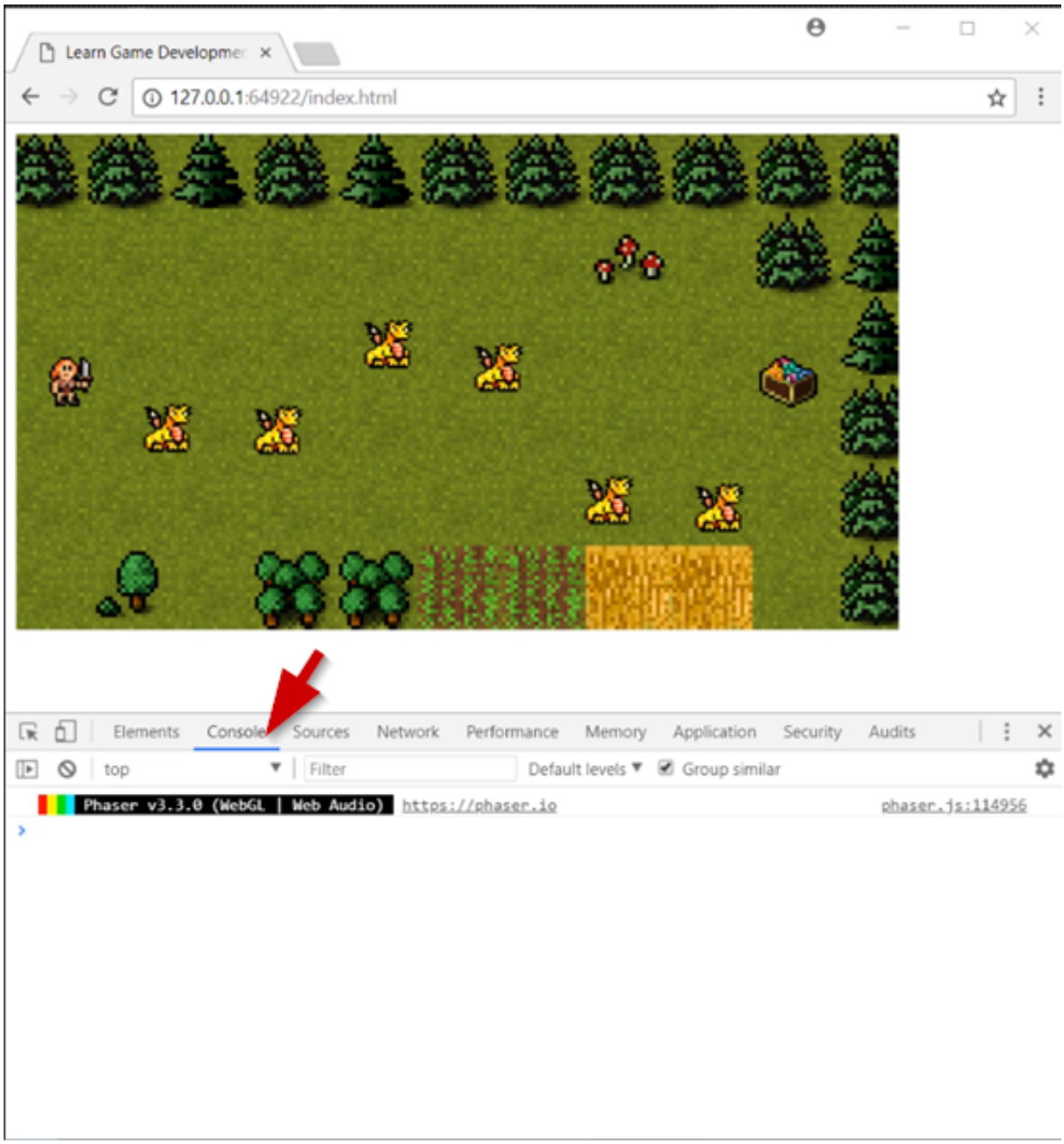
You can **download** Brackets from here: http://brackets.io

The **direct link** to download Brackets is here: Download Brackets

Brackets comes with a built-in web server and it is free to use.

- Web Server-Brackets
  - Download and install Brackets (https://brackets.io)
  - File-Open Folder
  - Live preview

Always keep the Chrome Developer Tools open when working.



There is a more advanced way of using a web server and this involves using a called http server.

The first step is to install **Node JS,** and you can download Node JS from here: https://nodejs.org/en/

The direct link to **download** Node JS is here: Node JS Download

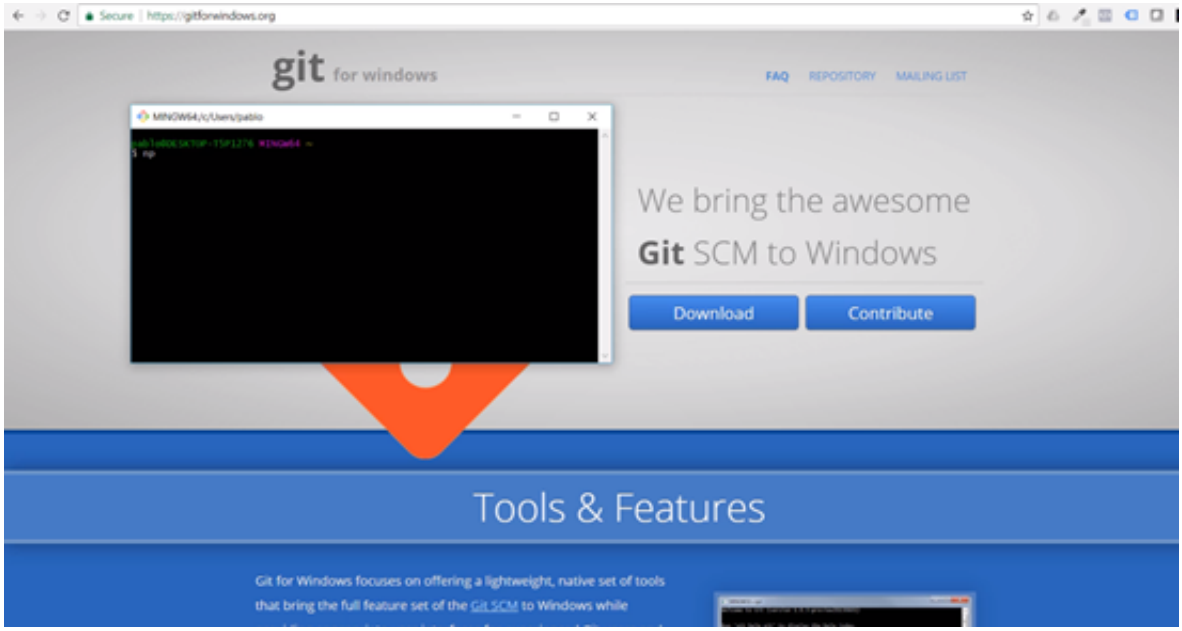
Node JS is an application that allows you to run JavaScript code on a server, in this case your computer.

For Window Users Only: Once you have downloaded Node JS you will need to download Git Bash, and this will give you access to a terminal on your system.

Mac users will already have access to the Mac terminal.

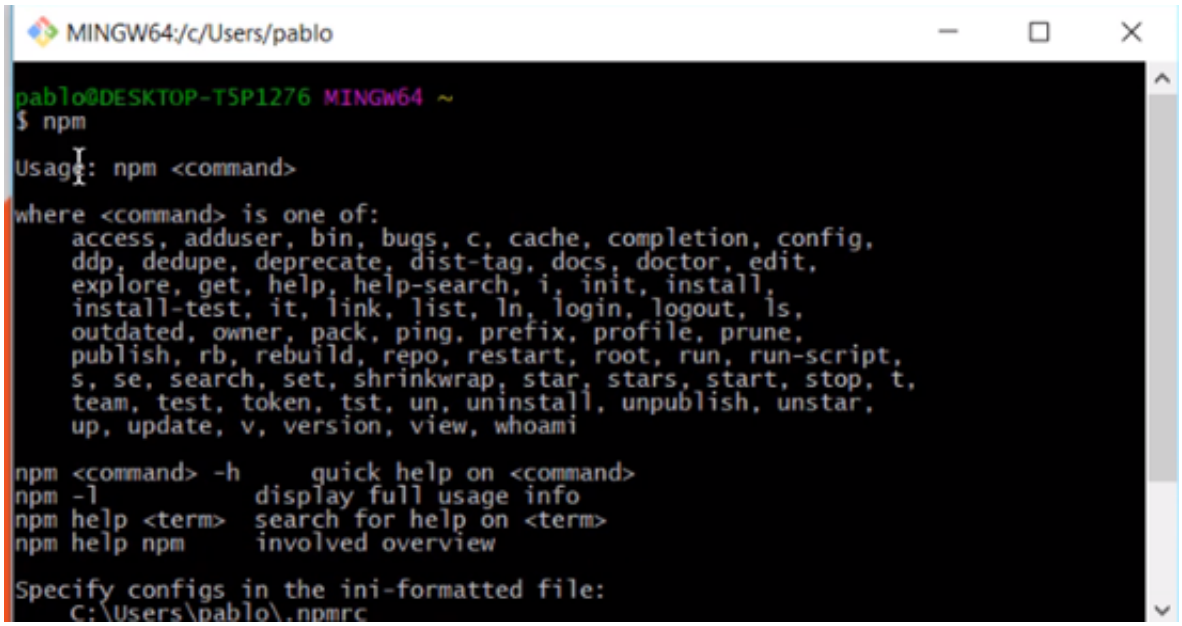
You can download **Git Bash** for windows here: https://gitforwindows.org/

The direct link to **download** Git Bash for windows is here: Git Bash



The first step is type **“npm”** in the terminal

If you type this and it is installed you will see this:



Once you verify that the npm is installed you can then type **“npm install http-server -g”** this is the name of the package that we want installed for us.

We now need to navigate and find our folder where the game is located.

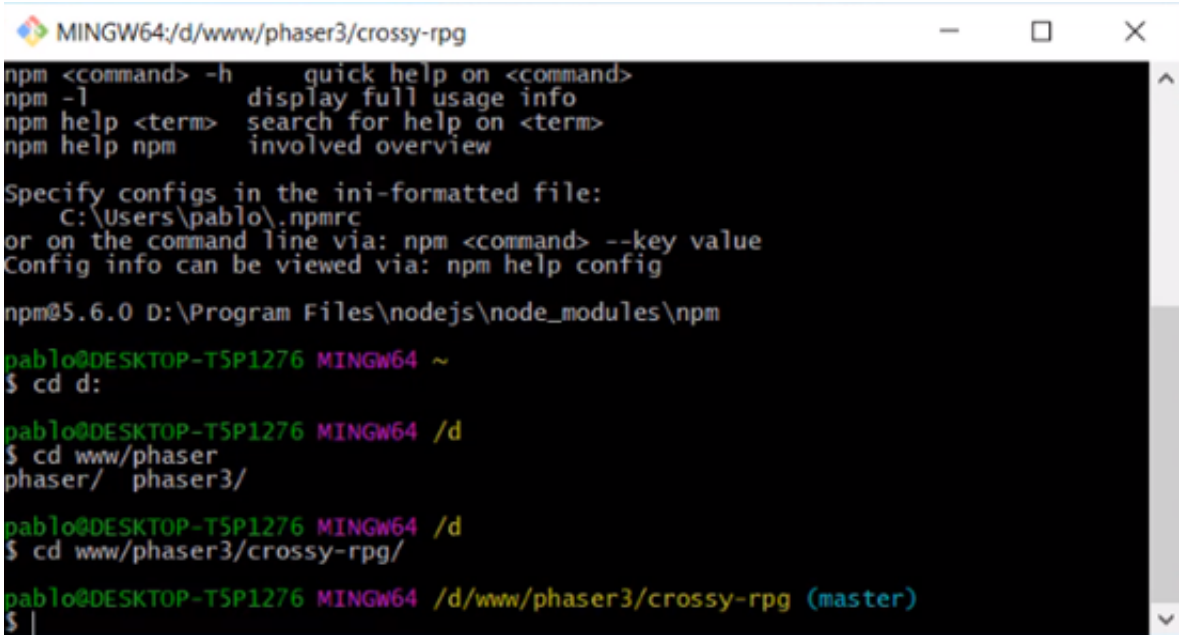
In my case the game is located in the D drive, the **www folder>Phaser3>crossy-rpg** this will be different in your case.

If you have issues with using the command line options here you can just use Brackets, Brackets is much easier to use and doesn’t require this extra setup for the http server setup.

So in the terminal you type **“cd d:”** this brings you to the D drive.

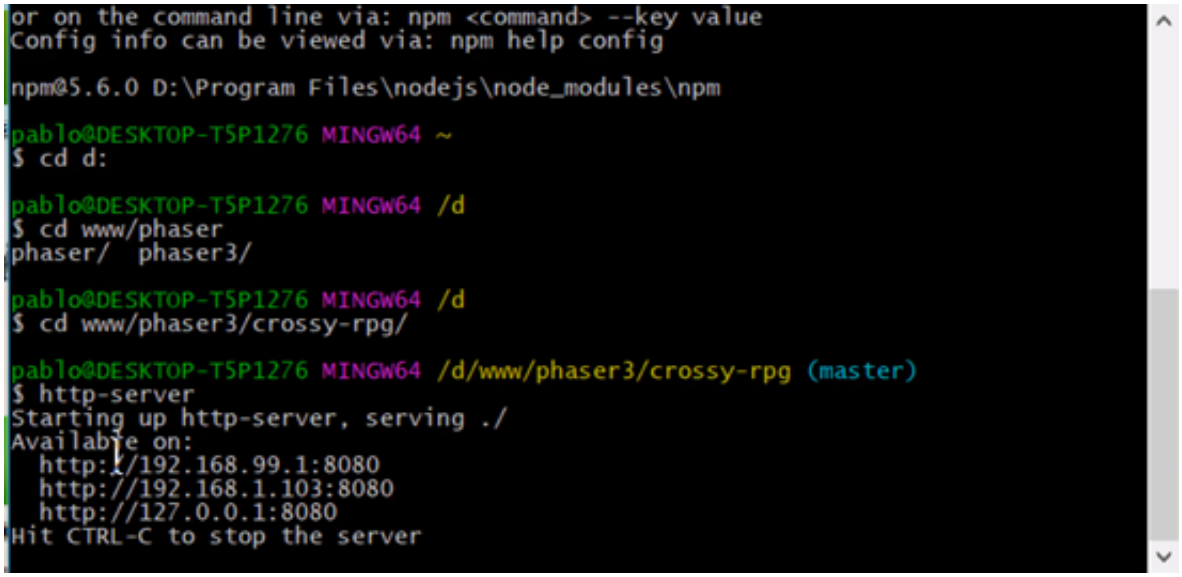
Then type **“cd www/phaser3/crossy-rpg/”**

Now you should be inside the correct folder in the terminal.

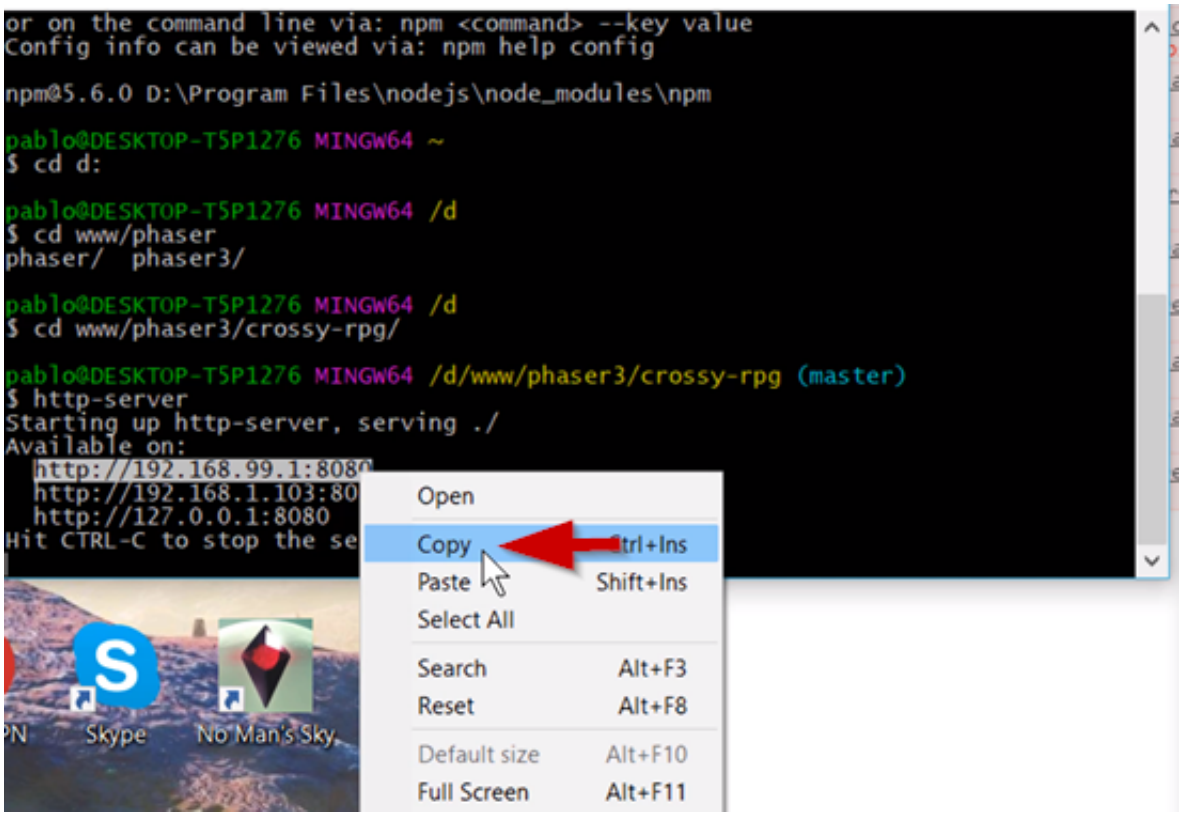


Now that you are inside the correct folder you then type **“http-server”** and that should launch the web server for that particular folder.

You should now see that the server is available in different URLs.



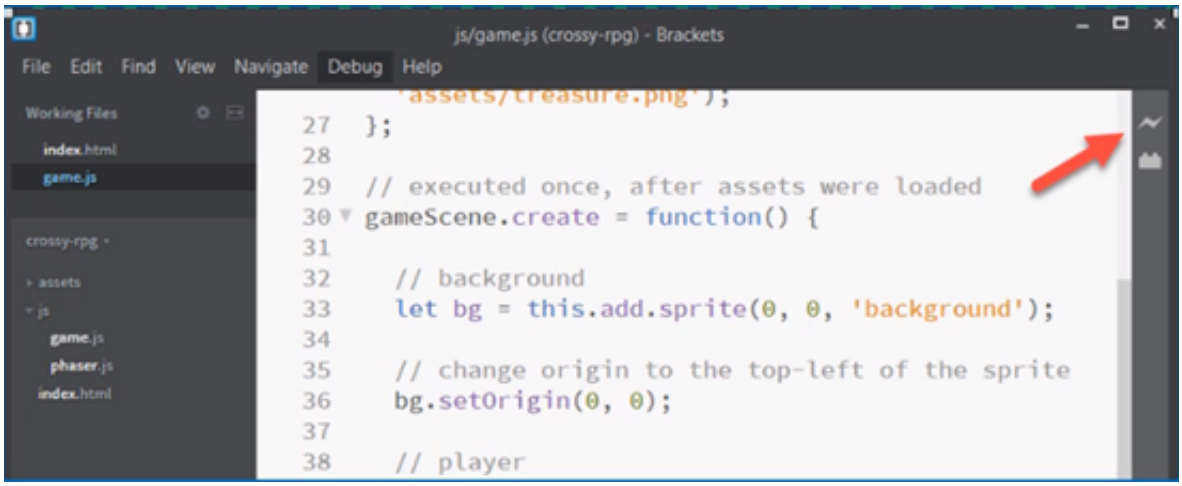
You then can copy one of the URLs and go back into Google Chrome or whatever web browser you are using and paste the copied code into the navigation bar.



And now the game will be working just like it should.

Learning Summary

- There are three basic requirements for Phaser development
    - Web browser
    - Code editor
    - Web Server
- A list of code editors that are free
    - Brackets(https://brackets.io)
    - Atom(https://atom.io)
    - Sublime text(https://www.sublimetext.com)
    - Visual Studio Code(https://code.visualstudio.com/)
- Web Server-Brackets
    - Download and install Brackets (https://brackets.io)
    - File-Open Folder
    - Live preview



- Web Server-http-server
    - Download and install Node.js (https://nodejs.org)
    - (Windows Only) install Git Bash (https:gitforwindows.org/)

- Open terminal, install http-server: npm install http-server -g
- Navigate to the project folder, run the server: http-server

Learning Goals

- Game Design Documents
- Importance of scoping
- Outlining game requirements

Game design documents(GDD's) are used in the game industry to **describe and communicate the concepts and requirements of a game** that will be developed.

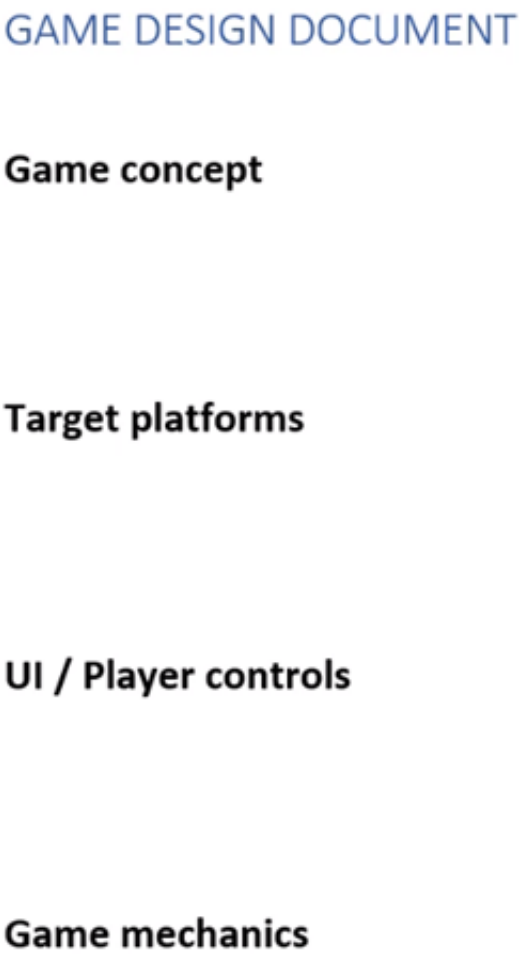
These documents can be very short or very long. The length will depend on the type of game and scope of the game as well.

The GDD's allow you to see everything in one place for the game that is being developed. It makes it easy for people who are developing a game on a team to get an idea of what the game should be, or other people who might just be interested in the development of the game. They can look at the GDD and see exactly what mechanics are going to be in the game and even see the type of art being used in the game.

GDD's can be used if you are working on a team or if you are just a solo developer.

You will want to include the **concept** of the game, the game **mechanics**, the **theme**, the **genre**, the **targeted platforms**, and any **art or sound assets** in the GDD.

A simple template for a GDD:



You can use just about word processing software to write the GDD in.

A GDD is considered to be a **“live”** document because it should be constantly updated and everyone working on the game should have access too the document at all times.

Learning Summary

- Game Design Documents (GDD)
    - Short documents to describe a game concept and requirements.
    - Used by and number of people (1 or many)
    - Includes:
        - Concept
        - Mechanics/UI
        - Platforms
        - Assets

GAME DESIGN DOCUMENT

Game concept

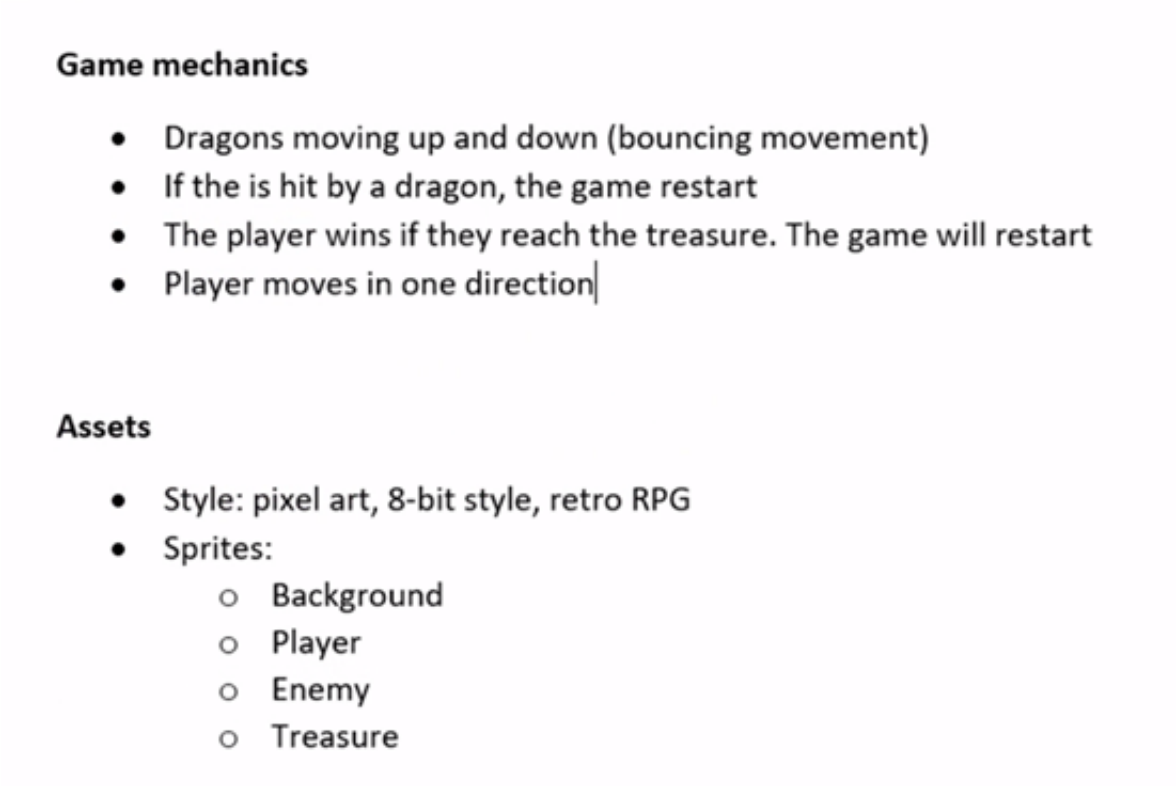
It's casual, 2D, "Frogger" style, where the player controls a Valkyrie who needs to reach a treasure and avoid dragons.

Target platforms

- Cross-platform: desktop and mobile
- Browser game

UI / Player controls

- Mouse click / touch screen
- When the player clicks/touches the screen, the character moves forward



Game mechanics

- Dragons moving up and down (bouncing movement)
- If the is hit by a dragon, the game restart
- The player wins if they reach the treasure. The game will restart
- Player moves in one direction

Assets

- Style: pixel art, 8-bit style, retro RPG
- Sprites:
  - Background
  - Player
  - Enemy
  - Treasure

- Challenge
  - Make a GDD of your game idea
  - Share it in the comments

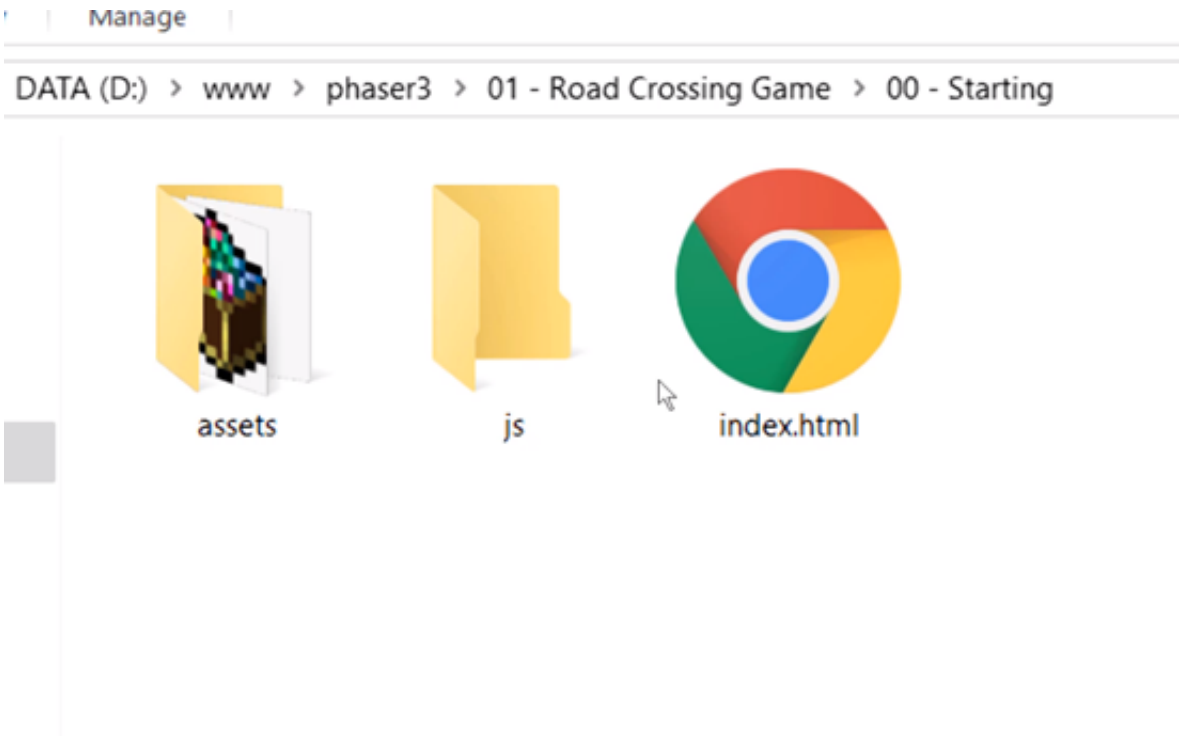
Learning Goals

- Including Phaser in a project.
- Passing configuration to a new game.
- Creating a Phaser game with a scene.

In this lesson we will begin creating the new Phaser game.

It will be an empty game on an empty scene, and you will learn what a scene is.

Download the files for the course, and make sure you find the folder named **“oo-Starting”**



You will be building the game along as you watch the lessons in the course.

The **js** folder is empty and this is where all the **JavaScript files** will go.

There is also a folder called **assets** and this contains all the images for the project.

You can **open the index.html file in your code editor**. I am using Atom in this course for my code editor.

This is a simple file that doesn’t really have anything.

We will need to **start by including the Phaser library.**

You can find the Phaser library at phaser.io.

The direct link is here: Phaser Download

If you want to put your game on the internet you will always go with the **minified version**. This version has all the comments stripped out of it.

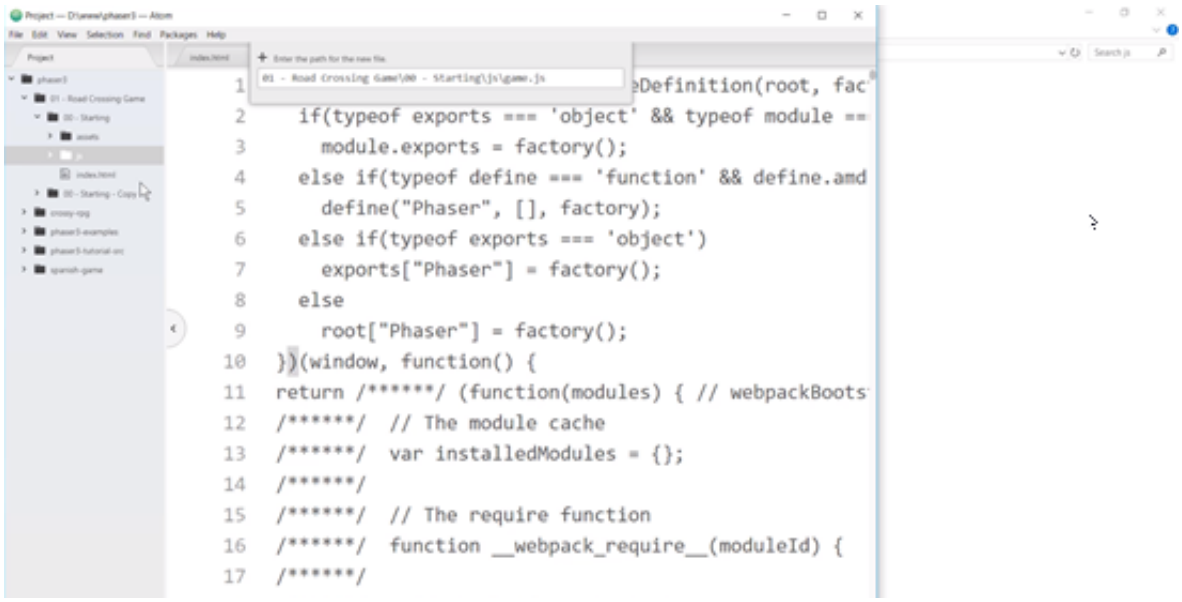
You need to **save the min.js file** in the JavaScript folder.

Then you need include the file in the index.html file.

See the code below and follow along:

```
<!doctype html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <title>Learn Game Development at Zenva.com</title>
</head>
<body>
  <script src="js/phaser.js"></script>
</body>
</html>
```


Create a new file called **“game.js”** and this is where the game code will go.



Make sure to **include that file in the index.html file**, but after the Phaser file. See the code below and follow along:

```
<!doctype html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <title>Learn Game Development at Zenva.com</title>
</head>
<body>
  <script src="js/phaser.js"></script>
  <script src="js/game.js"></script>
</body>
</html>
```

Inside the **game.js file** there are three things that need to happen. We need to **create a new scene**, set the **configuration of the game**, and **create the new game** and pass the **configuration** to it.

A **scene** is where all the action takes place. This is where all the game characters, enemies, and everything in the game will go in the scene. In Phaser you can actually have multiple scenes. You can use scenes to represent different screens, or in some cases different levels.

See the code below and follow along:

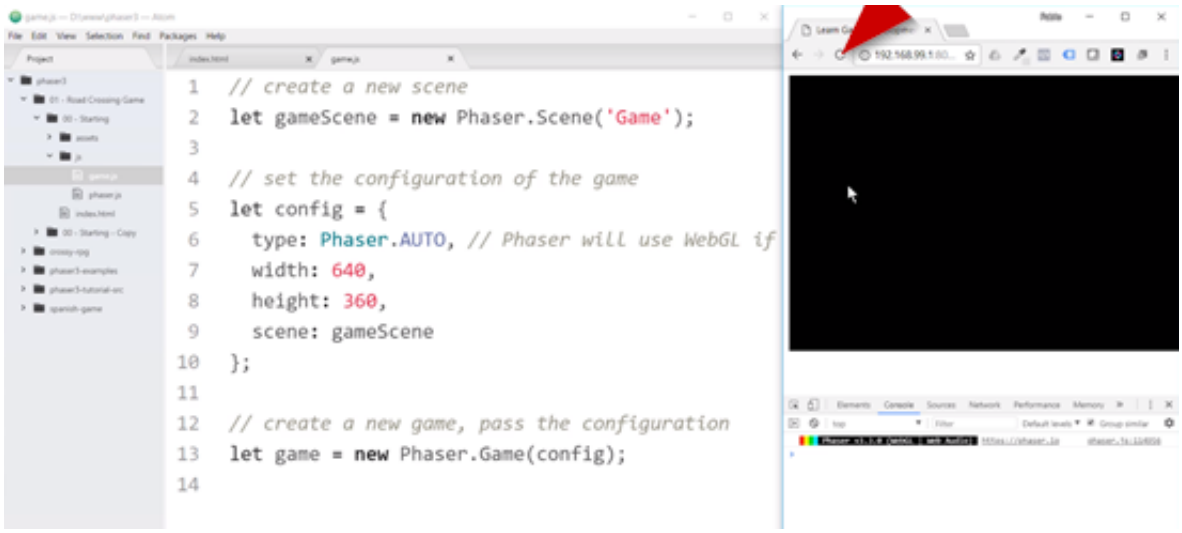
```
// create a new scene
let gameScene = new Phaser.Scene('Game');

// set the configuration of the game
let config = {
  type: Phaser.AUTO, // Phaser will use WebGL if available, if not it will use Canvas
  width: 640,
```

```
  height: 360,
  scene: gameScene
};

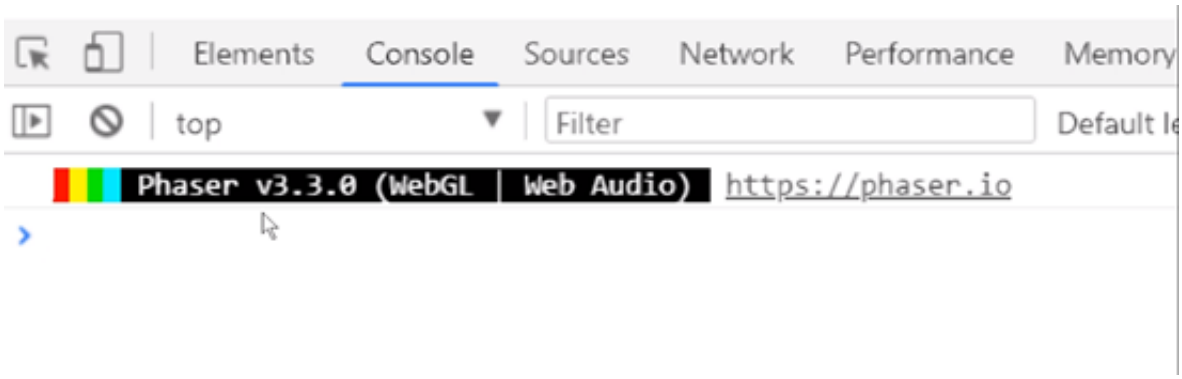
// create a new game, pass the configuration
let game = new Phaser.Game(config);
```

Then test out the changes by **refreshing** the web page:



There isn't much happening currently, there is a black screen of the size that we defined.

You will see that there is an indication in the **Console window** that Phaser is indeed working though.



In the next lesson we will be adding images to the screen.

Learning Summary

- Obtaining Phaser
  - Phaser homepage(http://phaser.io)
  - Version: 3.x
  - Options:
    - Direct Download

        - CDN
        - Github
        - npm
- Creating a new game
    - Create a scene
    - Configuration
    - Create game object

```
// create a new scene
let gameScene = new Phaser.Scene('Game');

// set the configuration of the game
let config = {
  type: Phaser.AUTO, // Phaser will use WebGL if available
  width: 640,
  height: 360,
  scene: gameScene
};

// create a new game, pass the configuration
let game = new Phaser.Game(config);
```

Corrections

At 11’00”, I say “340” when I should have said “640” (this was coded correctly on screen).

At 15’14” on the Slide “Coordinate System”, the text should instead read:
X coordinate: positive to the LEFT, negative to the RIGHT
Y coordinate: positive DOWNWARDS, negative UPWARDS

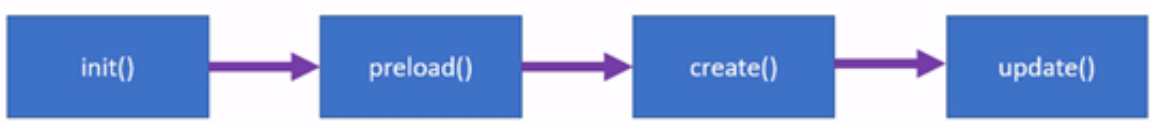
Learning Goals

- Phaser’s preload and create methods.
- Rendering a sprite.
- Coordinate system.
- Sprite origin.

In this lesson you will learn how to display images on the screen using Phaser. The images that will be used during this lesson can be downloaded from the course. They are in the assets folder. There are a total of 4 images in the folder: background, dragon, player, and treasure.

We will begin by loading the background, and the loading will occur inside of our scene.

The diagram for the **life cycle of a scene**:



The **init() method**– called once this is used to initiate certain parameters of your scene. This method is not always used.

preload() method-all asset preloading takes place. This means that Phaser will load all of the images, all the audio files, and other external files you may want to use in your game. All of the files will be loaded to memory so that they can be used without any delay. When you want to show an enemy or character, you don’t want the player to have to wait until that image is loaded.

create() method-called once and this is actually where you create the sprites and display them on the screen.

update() method-we will look at this method later on in the course, but it is called on each frame during game play. This method is used when things need to be checked all the time.

We will now create the preload method for the scene. See the code below and follow along:

```
// create a new scene
let gameScene = new Phaser.Scene('Game');

// load assets
gameScene.preload = function(){
  // load images
  this.load.image('background', 'assets/background.png');
  this.load.image('player', 'assets/player.png');
};
```



```
// called once after the preload ends
gameScene.create = function() {
  // create bg sprite
  let bg = this.add.sprite(0, 0, 'background');

  // change the origin to the top-left corner
  //bg.setOrigin(0,0);

  // place sprite in the center
  bg.setPosition(640/2, 360/2);

  let gameW = this.sys.game.config.width;
  let gameH = this.sys.game.config.height;

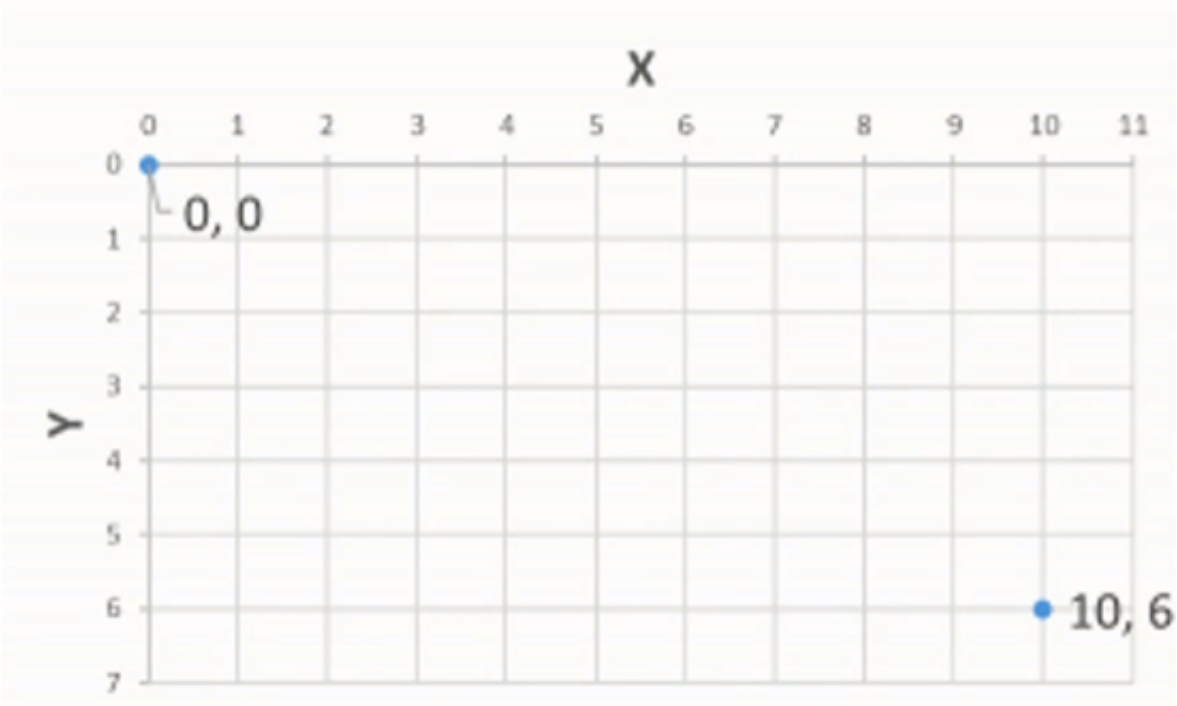
  console.log(gameW, gameH);

  console.log(bg);
  console.log(this);
};

// set the configuration of the game
let config = {
  type: Phaser.AUTO, // Phaser will use WebGL if available, if not it will use Canvas
  width: 640,
  height: 360,
  scene: gameScene
};

// create a new game, pass the configuration
let game = new Phaser.Game(config);
```

In Phaser games a **two dimensional coordinate system** is used. There is an X axis that goes from left to right. There is a Y axis that goes from top to bottom. The **origin(0,0)** is always at the top left corner of the game.



If we said we wanted the player to begin in position (5,4) What we are saying is that the center of our sprite will go in that position.

- **X:** Positive to the right, negative to the left.
- **Y:** Positive upwards, negative downwards.

By default the origin is the middle point on X and the middle point on Y.

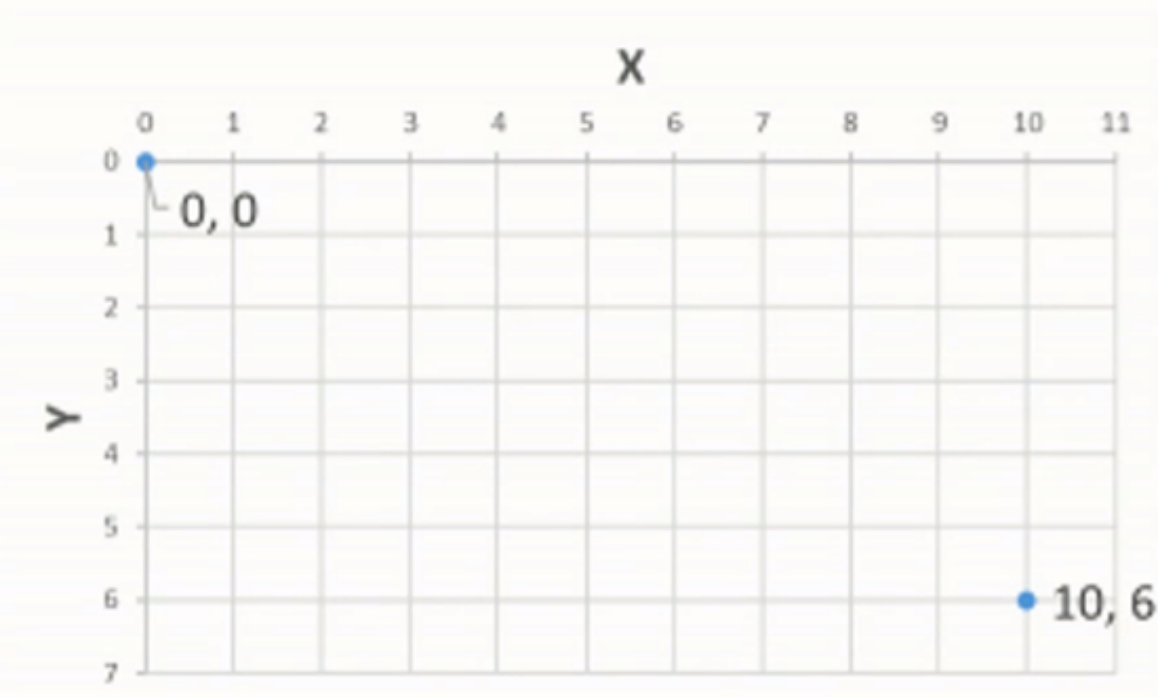
You can use the **bg.setOrigin(0,0)** method to change the origin of sprites.

Learning Summary

Preloading assets

```
// load assets
gameScene.preload = function(){
  // load images
  this.load.image('background', 'assets/background.png');
  this.load.image('player', 'assets/player.png');
};
```

- Sprites
  - Game characters and elements.
  - Origin: by default, the middle.
  - Position on x, y can be accessed and changed: set position.
- Coordinate system
  - X: Positive to the right, negative to the left.
  - Y: Positive upwards, negative downwards.



- Challenge
  - Place the sprite in the middle of the screen by changing the position.
  - Don’t change the origin.
  - Use the setPostion method.

Some important corrections to the content of this video:

1) Scaling a sprite by 2 does NOT double its size, it doubles it's width and it's height. If you double both the width and the height, the area after scaling is 4x the initial area!

2) Similarly, if you scale a sprite by 0.5 you are NOT reducing it's area by half — you are reducing both the width and the height by half. This means, the total area is multiplied by 0.25, so a quarter of the original area!

Learning Goals

- Sprite depth
- Changing the size of a sprite.
- Flipping a sprite on X or Y.

We are now familiar with how to position sprites on the screen, as we learned that in the previous lesson.

In this lesson you will learn how to increase or reduce the size of our sprites. This is called **scaling**. You will also learn how to flip sprites, and also what the depth of the sprite does when you render multiple sprites.

We will begin coding by adding a second sprite, which will be the player. See the code below and follow along:

```
// create a new scene
let gameScene = new Phaser.Scene('Game');

// load assets
gameScene.preload = function(){
  // load images
  this.load.image('background', 'assets/background.png');
  this.load.image('player', 'assets/player.png');
  this.load.image('enemy', 'assets/dragon.png');
};

// called once after the preload ends
gameScene.create = function() {
  // create bg sprite
  let bg = this.add.sprite(0, 0, 'background');

  // change the origin to the top-left corner
  bg.setOrigin(0,0);

  // create the player
  let player = this.add.sprite(70, 180, 'player');

  // we are reducing the width by 50%, and we are doubling the height
  player.setScale(0.5);

  // create an enemy
  let enemy1 = this.add.sprite(250, 180, 'enemy');
```



```
  enemy1.scaleX = 2;
  enemy1.scaleY = 2;

  // create a second enemy
  let enemy2 = this.add.sprite(450, 180, 'enemy');
  enemy2.displayWidth = 300;

  // flip
  enemy1.flipX = true;
  enemy1.flipY = true;
};

// set the configuration of the game
let config = {
  type: Phaser.AUTO, // Phaser will use WebGL if available, if not it will use Canvas
  width: 640,
  height: 360,
  scene: gameScene
};

// create a new game, pass the configuration
let game = new Phaser.Game(config);
```

By default sprites are shown in order, this order is in which they are added. Since we added the background first and then added the player next. The player is showing on top of the background. You can change this by changing the depth of the sprites. The higher the number, the more on top the sprite will be.

When you have sprites and you want to have control over what goes on top you can use the **depth property**.

In order to scale sprites you can use the **set scale method** which works in a similar way to set origin. This means we will have a value for both X and Y.Increase (>1) or reduce (<1) the size of sprites, on X and/or Y. displayWidth and displayHeight: rendererd size of the sprite.

```
// create the player
let player = this.add.sprite(70, 180, 'player');

// we are reducing the width by 50%, and we are doubl
player.setScale(0.5, 2);

// create an enemy
let enemy1 = this.add.sprite(250, 180, 'enemy');

enemy1.scaleX = 2;
enemy1.scaleY = 2;

// create a second enemy
let enemy2 = this.add.sprite(450, 180, 'enemy');
enemy2.displayWidth = 300;
```

You can **flip a sprite** for both X and Y axes.

```
// flip
enemy1.flipX = true;
enemy1.flipY = true;
```

Learning Summary

- Sprite depth
  - by default, sprites are stacked in the order that are created.
  - The depth property can be used to customize which ones go on top.
  - Sprites with a higher depth value are shown above those with a lower value.
- Scaling
  - Increase (>1) or reduce (<1) the size of sprites, on X and/or Y.
  - displayWidth and displayHeight: rendererd size of the sprite.

```
// create the player
let player = this.add.sprite(70, 180, 'player');

// we are reducing the width by 50%, and we are doubl
player.setScale(0.5, 2);

// create an enemy
let enemy1 = this.add.sprite(250, 180, 'enemy');

enemy1.scaleX = 2;
enemy1.scaleY = 2;

// create a second enemy
let enemy2 = this.add.sprite(450, 180, 'enemy');
enemy2.displayWidth = 300;
```

- Flipping
    - Can be done for both X and Y.

```
// flip
enemy1.flipX = true;
enemy1.flipY = true;
```

Learning Goals

- Rotating a sprite in angles and radians.
- Update method.
- Changing a property on each frame.

In this lesson you will learn about the **rotation of sprites and the update method**. The update method is part of the scenes life cycle and it is a method that we can use to change the property of something over time.

There are two ways of rotating a sprite. The rotation always takes place about its origins, so the position of the origin doesn’t change.

We will begin by increasing the size of the enemy a little bit, see the code below and follow along:

```
// create an enemy
  this.enemy1 = this.add.sprite(250, 180, 'enemy');
  this.enemy1.setScale(3);

  this.enemy1.angle = 45;
  this.enemy1.setAngle(45);
```

This is how you rotate a sprite by **picking the angles option**.

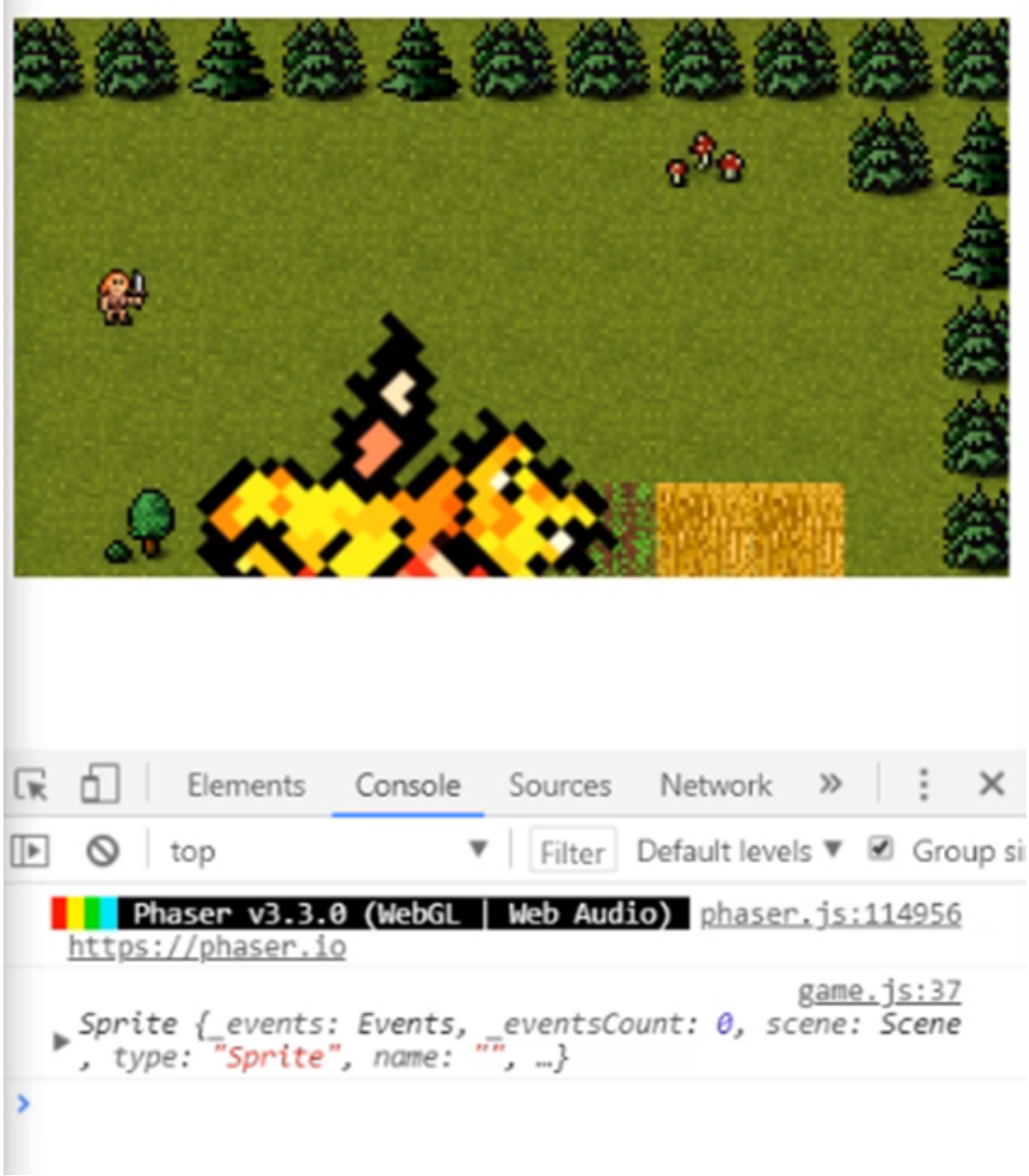
See the code below and follow along with **how to use the radians option to rotate the sprite**:

```
this.enemy1.rotation = Math.PI / 4;
this.enemy1.setRotation(Math.PI / 4);
```

You can see that we are rotating about the origin, which is in the middle.

Now if we didn't want to rotate about the middle anymore, and rotate about a specific point we could do that, see the code below and follow along:

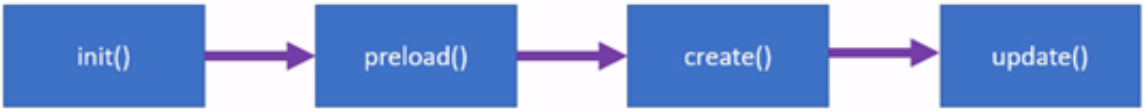
```
this.enemy1.setOrigin(0, 0);
this.enemy1.rotation = Math.PI / 4;
this.enemy1.setRotation(Math.PI / 4);
```



This is how you can rotate sprites and now you know how to change a sprites position, how to scale them up and down, flip sprites, and you now know how to rotate them.

The next thing to cover in this lesson is the **update method**.

In a previous lesson we went over how scenes had special methods.



The **update() method** is called 60 times per second if that is of course supported by the device.

60 frames per second(fps) is the frame rate that the framework aims to have.

We will now create the update method, see the code below and follow along:

```
// this is called up to 60 times per second
gameScene.update = function(){
  //this.enemy1.x += 1;

  this.enemy1.angle += 1;
```

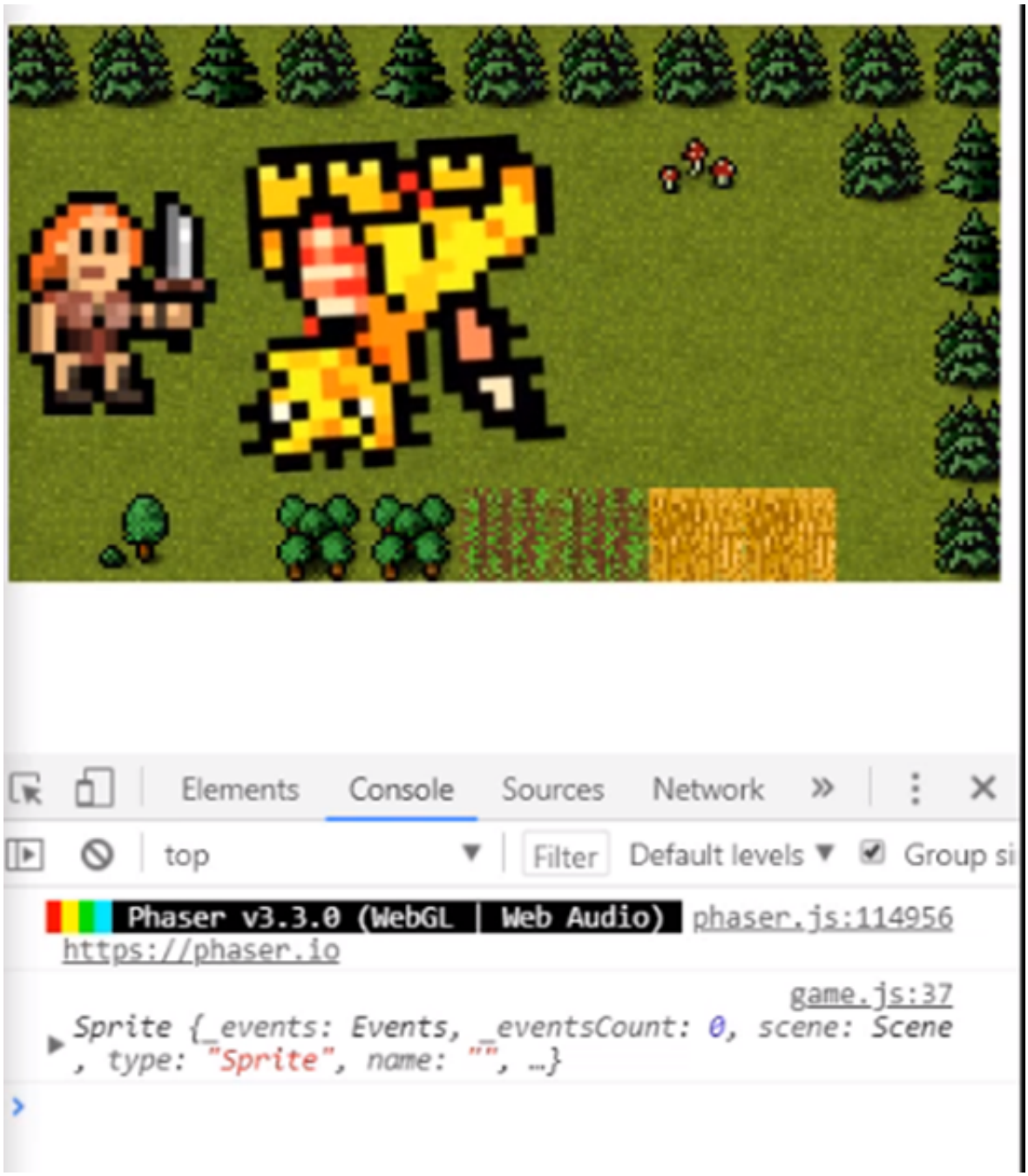
Now there is a challenge for you, and it has to do with playing with the update method.

Challenge– Make a sprite grow over time, only until it reaches twice it’s original dimensions(width and height)

Try to solve the challenge on your own without looking at the solution, but if you need help feel free to watch the solution and follow along.

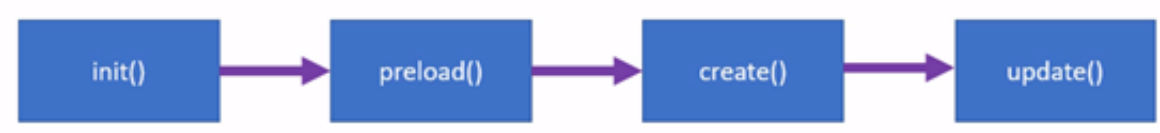
Solving the challenge:

```
// check if we've reached scale of 2
  if(this.player.scaleX < 2) {
    // make the player grow
    this.player.scaleX += 0.01;
    this.player.scaleY += 0.01;
  }
```

Learning Summary

- Rotation
  - Rotation using your left hand is positive.
  - In angles: mySprite.angle = 45;
  - In radians:mySprite.rotation = Math.PI/4;



- update()
    - Called once up to 60 times per second.
    - Used for things that need to be done or checked “at all times”
    - Executed for the first time after create()
- Challenge
    - Make a sprite grow over time, only until it reaches twice it’s original dimensions(width and height)

Learning Goals

- Detecting user input.
- Moving the player.
- Scene init method.
- Checking if two sprites overlap.
- Restarting a scene.

It is now time to implement the logic of the player. We will set up the ability for the player to move in this lesson. The player will be able to **move in the positive X direction**, and we will be adding the goal into the game as well. The goal will be at the end for the player to reach. The goal will be represented by the treasure chest image and then the player reaches the treasure chest we will restart the scene.

There was some housekeeping done to the **game.js file**. Everything that was inside the update method was deleted. The enemy was also deleted.

All you should have in the game to move forward is the player positioned in the middle of the screen.

See the screenshot below:

We want to **detect input in order to make the player move in the game**. There are a few built in ways to do this in Phaser 3.

See the code below and follow along:

```
// initiate scene parameters
gameScene.init = function() {
  // player speed
  this.playerSpeed = 5;
};

 // check for active input (left click / touch)
```



```
  if(this.input.activePointer.isDown) {
    // player walks
    this.player.x += this.playerSpeed;
  }
```

Refresh the page and test out the changes.

The player is moving quite fast actually. We can change the **playerSpeed from 5 to 3**.

```
// initiate scene parameters
gameScene.init = function() {
  // player speed
  this.playerSpeed = 3;
};
```

We now have the player moving and moving at a decent speed.

What we want to do now is add in the treasure chest to the game. We want to detect collision or overlapping of different elements when the player reaches that point.

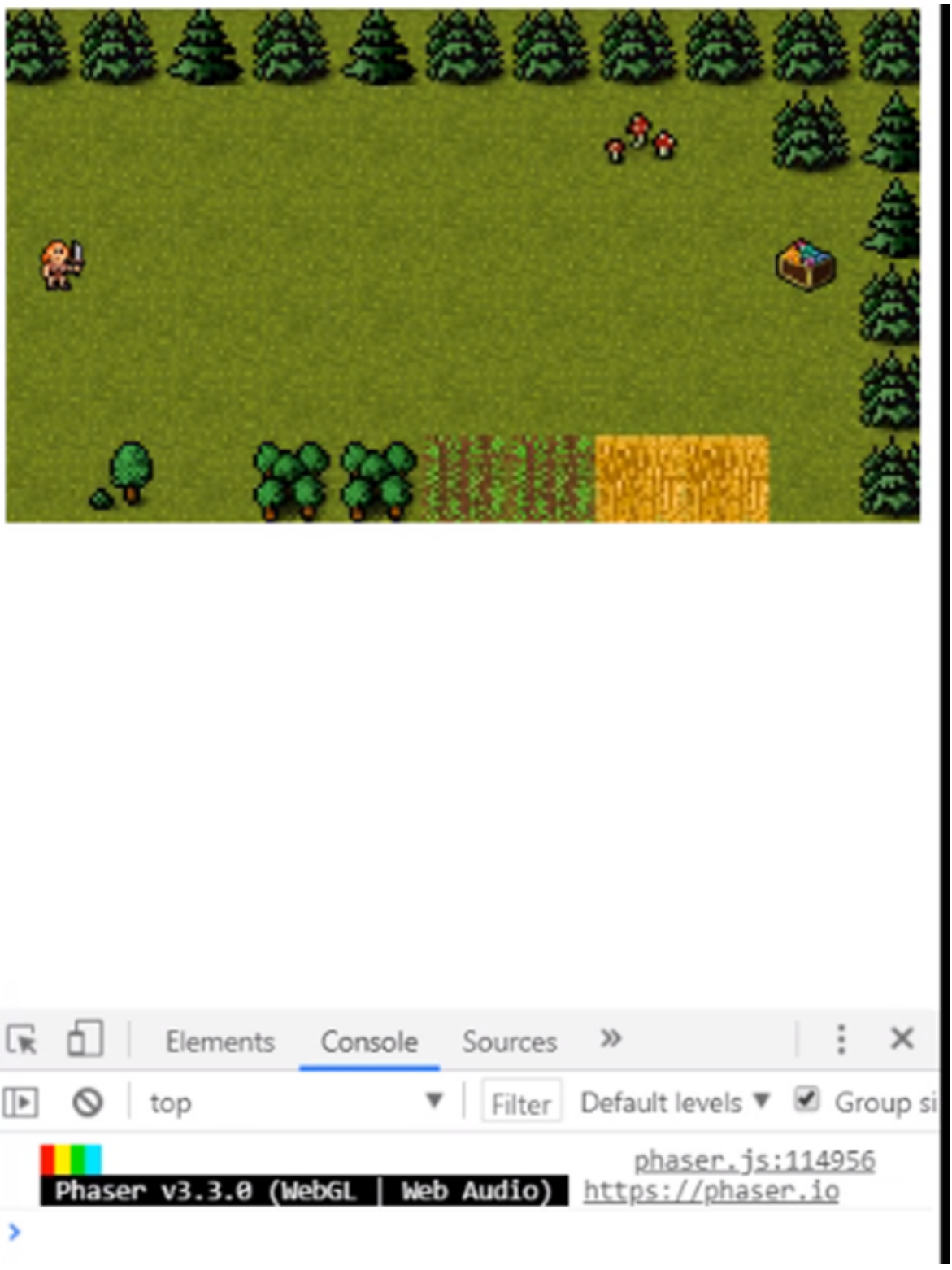
We need to first **pre-load** the sprite. See the code below and follow along:

```
// load assets
gameScene.preload = function(){
  // load images
  this.load.image('background', 'assets/background.png');
  this.load.image('player', 'assets/player.png');
  this.load.image('enemy', 'assets/dragon.png');
  this.load.image('goal', 'assets/treasure.png');
};
```

Now we need to **create a new sprite for that element** we just added above and change the scale of the treasure chest. See the code below and follow along:

```
// goal
  this.goal = this.add.sprite(this.sys.game.config.width - 80, this.sys.game.config.h
eight / 2, 'goal');
  this.goal.setScale(0.6);
```

Refresh the page.



What we want to do now is **check for collision with the player and the treasure chest.**

```
// treasure overlap check
  let playerRect = this.player.getBounds();
  let treasureRect = this.goal.getBounds();

  if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, treasureRect)) {
    console.log('reached goal!');
```



```
    // restart the Scene
    this.scene.restart();

    // make sure we leave this method
    return;
  }

};
```

Refresh the page and test out the changes.

The player should now move and when the player collides with the treasure chest the scene is restarted.

Here is the entire **game.js file**, if you are having issues you can compare your code to this:

```
// create a new scene
let gameScene = new Phaser.Scene('Game');

// initiate scene parameters
gameScene.init = function() {
  // player speed
  this.playerSpeed = 3;
};

// load assets
gameScene.preload = function(){
  // load images
  this.load.image('background', 'assets/background.png');
  this.load.image('player', 'assets/player.png');
  this.load.image('enemy', 'assets/dragon.png');
  this.load.image('goal', 'assets/treasure.png');
};

// called once after the preload ends
gameScene.create = function() {
  // create bg sprite
  let bg = this.add.sprite(0, 0, 'background');

  // change the origin to the top-left corner
  bg.setOrigin(0,0);

  // create the player
  this.player = this.add.sprite(40, this.sys.game.config.height / 2, 'player');

  // we are reducing the width and height by 50%
  this.player.setScale(0.5);

  // goal
  this.goal = this.add.sprite(this.sys.game.config.width - 80, this.sys.game.config.h
eight / 2, 'goal');
  this.goal.setScale(0.6);
```



```
};

// this is called up to 60 times per second
gameScene.update = function(){

  // check for active input (left click / touch)
  if(this.input.activePointer.isDown) {
    // player walks
    this.player.x += this.playerSpeed;
  }

  // treasure overlap check
  let playerRect = this.player.getBounds();
  let treasureRect = this.goal.getBounds();

  if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, treasureRect)) {
    console.log('reached goal!');

    // restart the Scene
    this.scene.restart();

    // make sure we leave this method
    return;
  }

};

// set the configuration of the game
let config = {
  type: Phaser.AUTO, // Phaser will use WebGL if available, if not it will use Canvas
  width: 640,
  height: 360,
  scene: gameScene
};

// create a new game, pass the configuration
let game = new Phaser.Game(config);
```

Learning Summary

- Detecting user input
  - There are many ways to do this in Phaser.
  - We are accessing the scene’s input object and check for the active input:

```
// check for active input (left click / touch)
if(this.input.activePointer.isDown) {
  // player walks
  this.player.x += this.playerSpeed;
}
```

- Init method
  - Used to set scene parameters, it runs before preload..

```
// initiate scene parameters
gameScene.init = function() {
  // player speed
  this.playerSpeed = 3;
};
```

- Checking if two sprites overlap
  - Geometrical utility to check if two rectangles overlap.
  - This can also be done using a physics engine (overkill for this game)

```
// treasure overlap check
let playerRect = this.player.getBounds();
let treasureRect = this.goal.getBounds();

if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, treasureRect)) {
  console.log('reached goal!');
```

- Restarting a scene
  - Tell the Scene Manager to reboot the current scene.

```
// restart the Scene
this.scene.manager.bootScene(this);
```


Learning Goals

- Creating a bouncing enemy.
- Assigning a random direction and speed.

We will be adding in a single enemy to our game that moves up and down.

We will **begin by adding the sprite for the dragon**. We already have the image loaded in the file. See the code below and follow along:

```
// enemy
  this.enemy = this.add.sprite(120, this.sys.game.config.height / 2, 'enemy');
  this.enemy.flipX = true;
  this.enemy.setScale(0.6);
```

We will now **make the enemy move up and down**. We will start by adding a speed for the enemy. See the code below and follow along:

```
// enemy speed
  this.enemySpeed = 3;

// enemy movement
  this.enemy.y += this.enemy.speed;
```

Refresh the page.

The enemy moves down, but not back up, and always at the same speed. The enemy isn't bouncing currently.

We can now **implement the bouncing speed**. See the code below and follow along:

```
// boundaries
 this.enemyMinY = 80;
 this.enemyMaxY = 280;

// check we haven't passed min or max Y
 let conditionUp = this.enemy.speed < 0 && this.enemy.y <= this.enemyMinY;
 let conditionDown = this.enemy.speed > 0 && this.enemy.y >= this.enemyMaxY;

 // if we passed the upper or lower limit, reverse
 if(conditionUp || conditionDown) {
   this.enemy.speed *= -1;
 }
```

We now have a bouncy enemy dragon and its always moving at the same speed. We will be adding in more enemy dragons in the next lesson, but we will not want them all moving at the same speed. We also don't want the dragons moving in the same direction at the same time, we want to add some randomization to the movement and speed to the enemy dragons.

We will now **setup the random enemy speed**. See the code below and follow along:

```
// enemy speed
  this.enemyMinSpeed = 2;
  this.enemyMaxSpeed = 4.5;

// set enemy speed
  let dir = Math.random() < 0.5 ? 1 : -1;
  let speed = this.enemyMinSpeed + Math.random() * (this.enemyMaxSpeed - this.enemyMi
nSpeed);
  this.enemy.speed = dir * speed;
```

Refresh the page a few times and you will see that the dragon moves at different speeds each time you refresh the page.

Here is the entire **game.js** file if you are having issues with your code you can compare it to this:

```
// create a new scene
let gameScene = new Phaser.Scene('Game');

// initiate scene parameters
gameScene.init = function() {
  // player speed
  this.playerSpeed = 3;

  // enemy speed
  this.enemyMinSpeed = 2;
  this.enemyMaxSpeed = 4.5;

  // boundaries
  this.enemyMinY = 80;
  this.enemyMaxY = 280;
};

// load assets
gameScene.preload = function(){
  // load images
  this.load.image('background', 'assets/background.png');
  this.load.image('player', 'assets/player.png');
  this.load.image('enemy', 'assets/dragon.png');
  this.load.image('goal', 'assets/treasure.png');
};

// called once after the preload ends
gameScene.create = function() {
  // create bg sprite
  let bg = this.add.sprite(0, 0, 'background');

  // change the origin to the top-left corner
  bg.setOrigin(0,0);

  // create the player
  this.player = this.add.sprite(40, this.sys.game.config.height / 2, 'player');
```



```
  // we are reducing the width and height by 50%
  this.player.setScale(0.5);

  // goal
  this.goal = this.add.sprite(this.sys.game.config.width - 80, this.sys.game.config.h
eight / 2, 'goal');
  this.goal.setScale(0.6);

  // enemy
  this.enemy = this.add.sprite(120, this.sys.game.config.height / 2, 'enemy');
  this.enemy.flipX = true;
  this.enemy.setScale(0.6);

  // set enemy speed
  let dir = Math.random() < 0.5 ? 1 : -1;
  let speed = this.enemyMinSpeed + Math.random() * (this.enemyMaxSpeed - this.enemyMi
nSpeed);
  this.enemy.speed = dir * speed;
};

// this is called up to 60 times per second
gameScene.update = function(){

  // check for active input (left click / touch)
  if(this.input.activePointer.isDown) {
    // player walks
    this.player.x += this.playerSpeed;
  }

  // treasure overlap check
  let playerRect = this.player.getBounds();
  let treasureRect = this.goal.getBounds();

  if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, treasureRect)) {
    console.log('reached goal!');

    // restart the Scene
    this.scene.restart();
    return;
  }

  // enemy movement
  this.enemy.y += this.enemy.speed;

  // check we haven't passed min or max Y
  let conditionUp = this.enemy.speed < 0 && this.enemy.y <= this.enemyMinY;
  let conditionDown = this.enemy.speed > 0 && this.enemy.y >= this.enemyMaxY;

  // if we passed the upper or lower limit, reverse
  if(conditionUp || conditionDown) {
    this.enemy.speed *= -1;
  }

};
```



```
// set the configuration of the game
let config = {
  type: Phaser.AUTO, // Phaser will use WebGL if available, if not it will use Canvas
  width: 640,
  height: 360,
  scene: gameScene
};

// create a new game, pass the configuration
let game = new Phaser.Game(config);
```

Learning Summary

- Bouncing movement
  - Move the enemy
  - Check if it's reached any of the limits.
  - If it's reached a limit, and it was moving in that direction, reverse speed.

```
// check we haven't passed min or max Y
let conditionUp = this.enemy.speed < 0 && this.enemy.y <= this.enemyMinY;
let conditionDown = this.enemy.speed > 0 && this.enemy.y >= this.enemyMaxY;
```

- Random velocity
  - Direction

```
let dir = Math.random() < 0.5 ? 1 : -1;
```

  - Speed

```
let speed = this.enemyMinSpeed + Math.random() * (this.enemyMaxSpeed - this.enemyMinSpeed);
```

Learning Goals

- Groups in Phaser
- Creating multiple sprites in a group.
- Applying a method to all group elements.
- Implementing the enemies group.

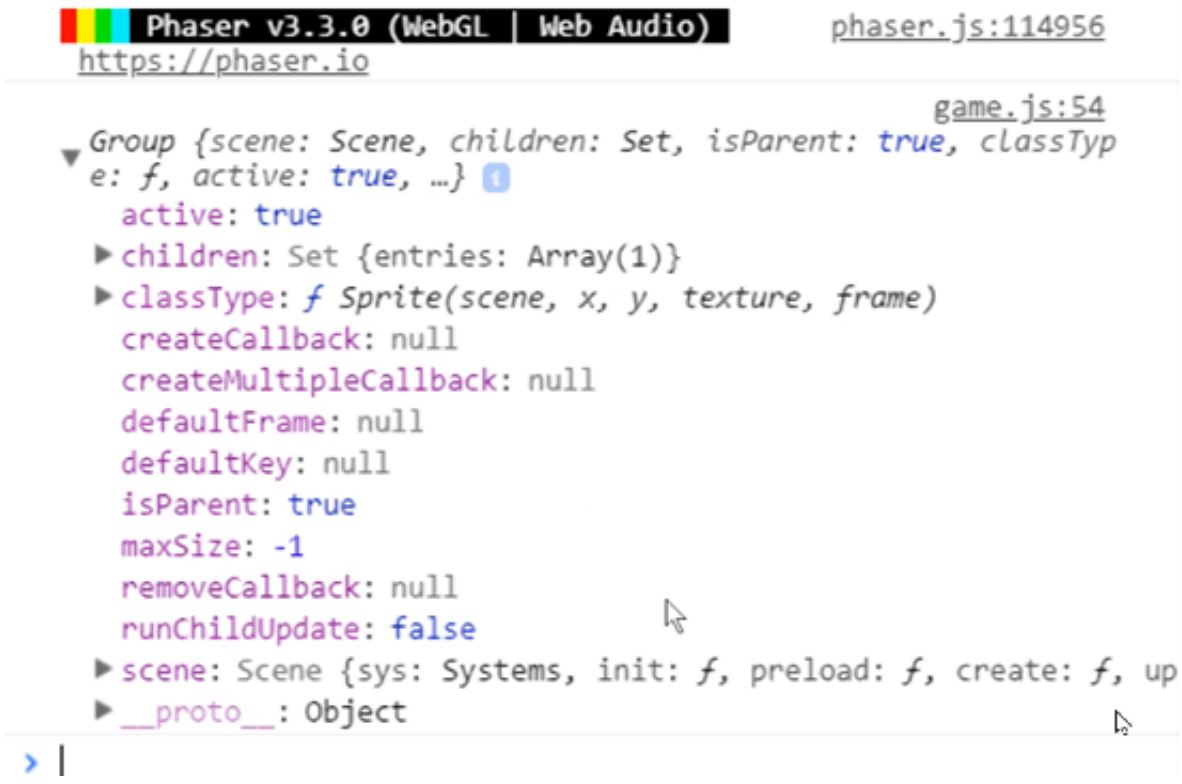
In this lesson you will be introduced to the concept of **groups**. You can think of a group as a collection of sprites.

We will begin by **creating a group for our enemies**. See the code below and follow along:

```
//enemy
this.enemies + this.add.group();

this.enemy = this.add.sprite(120, this.sys.game.config.height / 2, 'enemy');
this.enemy.flipX = true;
this.enemy.setScale(0.6);
```

Refresh the page and look at the Console window:



We will be able to access our elements at any time by simply doing the **get children method**.

See the code below and follow along:

```
console.log(this.enemies.getChildren());
```


You can see that you are given a normal array with all the sprite elements.

You can **use groups to change the scale of your elements.**

You can pass an object to the group declaration and that object can contain how many elements we want to create and what sprites are seen.

See the code below and follow along:

```
// enemy group
  this.enemies = this.add.group({
    key: 'enemy',
    repeat: 5,
    setXY: {
      x: 110,
      y: 100,
      stepX: 80,
      stepY: 20
    }
```

Refresh the page.

So we now have a bunch of sprites that are separated and the previous enemy is bouncing up and down.

Now we need to be able to move the other dragons and then flip them. See the code below and follow along:

```
// enemy group
  this.enemies = this.add.group({
    key: 'enemy',
    repeat: 5,
    setXY: {
      x: 90,
      y: 100,
      stepX: 80,
      stepY: 20
    }
  });

  // setting scale to all group elements
  Phaser.Actions.ScaleXY(this.enemies.getChildren(), -0.4, -0.4);

  // set flipX, and speed
  Phaser.Actions.Call(this.enemies.getChildren(), function(enemy){
    // flip enemy
    enemy.flipX = true;

    // set speed
    let dir = Math.random() < 0.5 ? 1 : -1;
    let speed = this.enemyMinSpeed + Math.random() * (this.enemyMaxSpeed - this.enemy
MinSpeed);
    enemy.speed = dir * speed;

  }, this);
};
```

Make sure you have deleted the previous created enemy.

We can now setup the movement. See the code below and follow along:

```
// get enemies
  let enemies = this.enemies.getChildren();
  let numEnemies = enemies.length;

  for(let i = 0; i< numEnemies; i++) {

    // enemy movement
    enemies[i].y += enemies[i].speed;

    // check we haven't passed min or max Y
    let conditionUp = enemies[i].speed < 0 && enemies[i].y <= this.enemyMinY;
    let conditionDown = enemies[i].speed > 0 && enemies[i].y >= this.enemyMaxY;

    // if we passed the upper or lower limit, reverse
    if(conditionUp || conditionDown) {
      enemies[i].speed *= -1;
    }
```


Refresh the page.

We now have all of our enemies moving at random speeds.

There is one thing missing though with the game and that is the collision checking with the enemies and the player.

See the code below and follow along:

```
// check enemy overlap
    let enemyRect = enemies[i].getBounds();

    if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, enemyRect)) {
      console.log('Game over!');

      // restart the Scene
      this.scene.restart();
      return;
    }
  }
```

Refresh the page and test out the changes. You will now see that the collision check is indeed working. Every time that player collides with an enemy the scene is restarted.

Feel free to adapt the game to make it easier if you wish.

What we need to do in the next lesson is take care of the game over implementation.

Here is the update **game.js** file for you to compare yours to if you are having issues:


```
// create a new scene
let gameScene = new Phaser.Scene('Game');

// initiate scene parameters
gameScene.init = function() {
  // player speed
  this.playerSpeed = 3;

  // enemy speed
  this.enemyMinSpeed = 2;
  this.enemyMaxSpeed = 4.5;

  // boundaries
  this.enemyMinY = 80;
  this.enemyMaxY = 280;
};

// load assets
gameScene.preload = function(){
  // load images
  this.load.image('background', 'assets/background.png');
  this.load.image('player', 'assets/player.png');
  this.load.image('enemy', 'assets/dragon.png');
  this.load.image('goal', 'assets/treasure.png');
};

// called once after the preload ends
gameScene.create = function() {
  // create bg sprite
  let bg = this.add.sprite(0, 0, 'background');

  // change the origin to the top-left corner
  bg.setOrigin(0,0);

  // create the player
  this.player = this.add.sprite(40, this.sys.game.config.height / 2, 'player');

  // we are reducing the width and height by 50%
  this.player.setScale(0.5);

  // goal
  this.goal = this.add.sprite(this.sys.game.config.width - 80, this.sys.game.config.h
eight / 2, 'goal');
  this.goal.setScale(0.6);

  // enemy group
  this.enemies = this.add.group({
    key: 'enemy',
    repeat: 5,
    setXY: {
      x: 90,
      y: 100,
      stepX: 80,
      stepY: 20
    }
```

```
  });

  // setting scale to all group elements
  Phaser.Actions.ScaleXY(this.enemies.getChildren(), -0.4, -0.4);

  // set flipX, and speed
  Phaser.Actions.Call(this.enemies.getChildren(), function(enemy){
    // flip enemy
    enemy.flipX = true;

    // set speed
    let dir = Math.random() < 0.5 ? 1 : -1;
    let speed = this.enemyMinSpeed + Math.random() * (this.enemyMaxSpeed - this.enemy
MinSpeed);
    enemy.speed = dir * speed;

  }, this);
};

// this is called up to 60 times per second
gameScene.update = function(){

  // check for active input (left click / touch)
  if(this.input.activePointer.isDown) {
    // player walks
    this.player.x += this.playerSpeed;
  }

  // treasure overlap check
  let playerRect = this.player.getBounds();
  let treasureRect = this.goal.getBounds();

  if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, treasureRect)) {
    console.log('reached goal!');

    // restart the Scene
    this.scene.restart();
    return;
  }

  // get enemies
  let enemies = this.enemies.getChildren();
  let numEnemies = enemies.length;

  for(let i = 0; i< numEnemies; i++) {

    // enemy movement
    enemies[i].y += enemies[i].speed;

    // check we haven't passed min or max Y
    let conditionUp = enemies[i].speed < 0 && enemies[i].y <= this.enemyMinY;
    let conditionDown = enemies[i].speed > 0 && enemies[i].y >= this.enemyMaxY;

    // if we passed the upper or lower limit, reverse
    if(conditionUp || conditionDown) {
```



```
      enemies[i].speed *= -1;
    }

    // check enemy overlap
    let enemyRect = enemies[i].getBounds();

    if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, enemyRect)) {
      console.log('Game over!');

      // restart the Scene
      this.scene.restart();
      return;
    }
  }



};

// set the configuration of the game
let config = {
  type: Phaser.AUTO, // Phaser will use WebGL if available, if not it will use Canvas
  width: 640,
  height: 360,
  scene: gameScene
};

// create a new game, pass the configuration
let game = new Phaser.Game(config);
```

Learning Summary

- Groups
    - Collection of sprites
    - Ability to create multiple sprites
    - Ability to apply specific methods to all sprites
    - Groups don’t have X, Y coordinates. Member sprites do


```
// enemy group
this.enemies = this.add.group({
    key: 'enemy',
    repeat: 5,
    setXY: {
        x: 90,
        y: 100,
        stepX: 80,
        stepY: 20
    }
});
```

- Calling a custom method

```
Phaser.Actions.Call(this.enemies.getChildren(), function(item){
    // method logic here

}, this);
```

Learning Goals

- Camera effects: Shake and fade
- Chaining effects
- Event listening

In this lesson we will be adding some nice **camera effects** to our game. We will be using events for this, and you will learn about how to use events in Phaser as well.

Currently all we are doing is restarting the scene once we collide with an enemy or the player reaches the treasure.

We need to move our restart scene code into a separate method. See the code below and follow along:

```
// end game
      return this.gameOver();
```

We can now being working on the camera shake effect. See the code below and follow along:

```
gameScene.gameOver = function() {

// shake camera
this.cameras.main.shake(500);
```

Refresh the page and test out the changes.

You should now see the camera shake effect when the player collides with an enemy. What we want to do now is restart the scene once the camera shake effect completes.

See the code below and follow along:

```
gameScene.gameOver = function() {

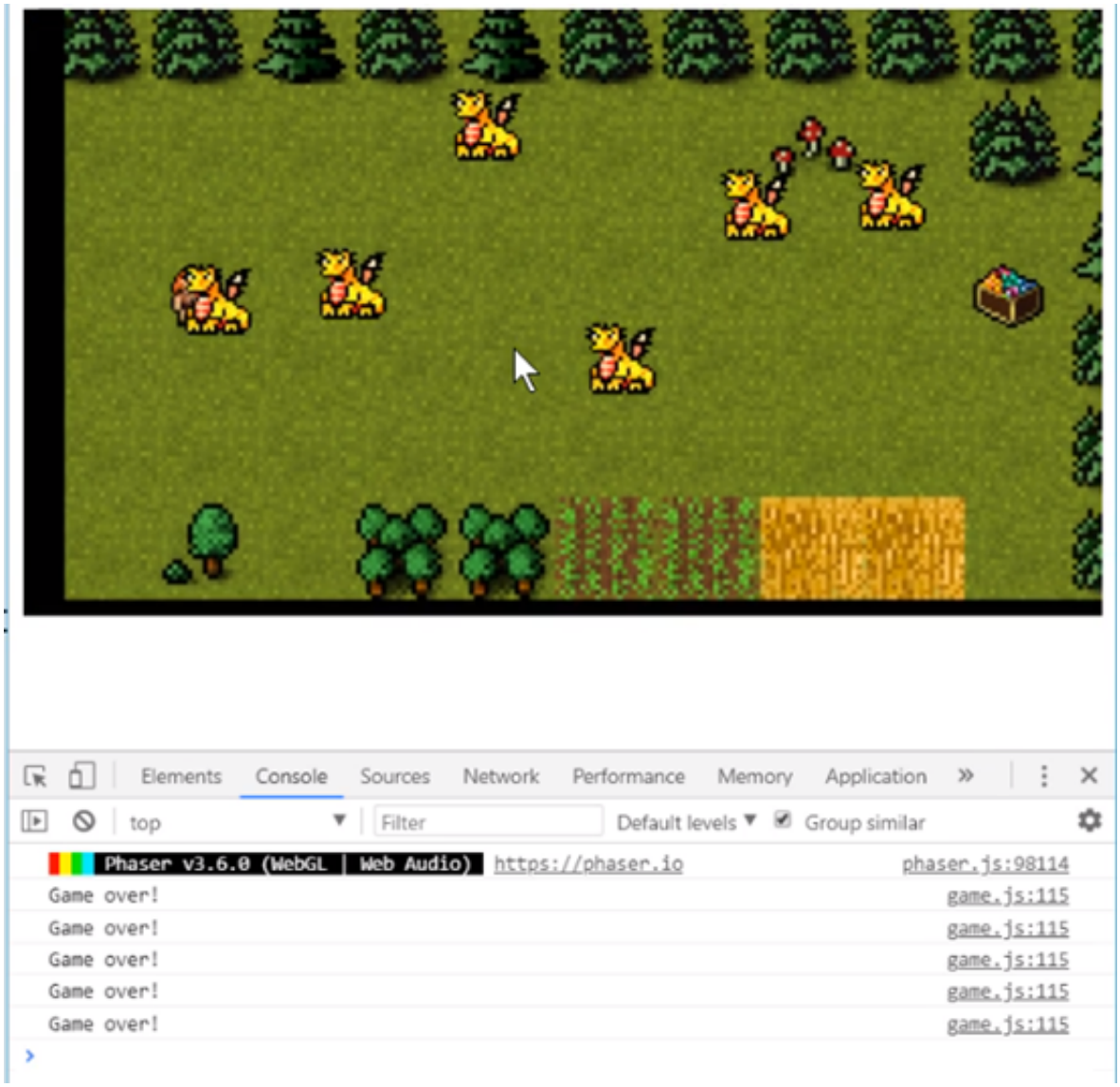
   // shake camera
  this.cameras.main.shake(500);

  // listen for event completion
  this.cameras.main.on('camerashakecomplete', function(camera, effect){

     // restart the Scene
    this.scene.restart();
  }, this);


};
```


Refresh the page and test the changes out.



So now when we collide with something the **Game Over** is being called but then while the camera event is taking place, we are calling the camera shake over and over again.

We don't really want this so we need to have some sort of flag that helps us call it just once.

We can create a **boolean variable** and take care of this pretty easily. See the code below and follow along:

```
 // we are not terminating
  this.isTerminating = false;

gameScene.gameOver = function() {

  // initiated game over sequence
  this.isTerminating = true;

  // shake camera
  this.cameras.main.shake(500);
```



```
  // listen for event completion
  this.cameras.main.on('camerashakecomplete', function(camera, effect){


};

// don't execute if we are terminating
  if(this.isTerminating) return;
```

Refresh the page and test the changes. You can now see that game over is happening once now.

We can now add the **camera fade** to the game. see the code below and follow along:

```
 // fade out
    this.cameras.main.fade(500);
  }, this);

  this.cameras.main.on('camerafadeoutcomplete', function(camera, effect){
    // restart the Scene
    this.scene.restart();
  }, this);
```

Refresh the page and you will see that once the camera shakes the scene will fade.

Here is the entire final version of the **game.js** file:

```
// create a new scene
let gameScene = new Phaser.Scene('Game');

// initiate scene parameters
gameScene.init = function() {
  // player speed
  this.playerSpeed = 3;

  // enemy speed
  this.enemyMinSpeed = 2;
  this.enemyMaxSpeed = 4.5;

  // boundaries
  this.enemyMinY = 80;
  this.enemyMaxY = 280;

  // we are not terminating
  this.isTerminating = false;
};

// load assets
gameScene.preload = function(){
```



```
  // load images
  this.load.image('background', 'assets/background.png');
  this.load.image('player', 'assets/player.png');
  this.load.image('enemy', 'assets/dragon.png');
  this.load.image('goal', 'assets/treasure.png');
};

// called once after the preload ends
gameScene.create = function() {
  // create bg sprite
  let bg = this.add.sprite(0, 0, 'background');

  // change the origin to the top-left corner
  bg.setOrigin(0,0);

  // create the player
  this.player = this.add.sprite(40, this.sys.game.config.height / 2, 'player');

  // we are reducing the width and height by 50%
  this.player.setScale(0.5);

  // goal
  this.goal = this.add.sprite(this.sys.game.config.width - 80, this.sys.game.config.h
eight / 2, 'goal');
  this.goal.setScale(0.6);

  // enemy group
  this.enemies = this.add.group({
    key: 'enemy',
    repeat: 5,
    setXY: {
      x: 90,
      y: 100,
      stepX: 80,
      stepY: 20
    }
  });

  // setting scale to all group elements
  Phaser.Actions.ScaleXY(this.enemies.getChildren(), -0.4, -0.4);

  // set flipX, and speed
  Phaser.Actions.Call(this.enemies.getChildren(), function(enemy){
    // flip enemy
    enemy.flipX = true;

    // set speed
    let dir = Math.random() < 0.5 ? 1 : -1;
    let speed = this.enemyMinSpeed + Math.random() * (this.enemyMaxSpeed - this.enemy
MinSpeed);
    enemy.speed = dir * speed;

  }, this);
};
```

```
// this is called up to 60 times per second
gameScene.update = function(){

  // don't execute if we are terminating
  if(this.isTerminating) return;

  // check for active input (left click / touch)
  if(this.input.activePointer.isDown) {
    // player walks
    this.player.x += this.playerSpeed;
  }

  // treasure overlap check
  let playerRect = this.player.getBounds();
  let treasureRect = this.goal.getBounds();

  if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, treasureRect)) {
    console.log('reached goal!');

    // end game
    return this.gameOver();
  }

  // get enemies
  let enemies = this.enemies.getChildren();
  let numEnemies = enemies.length;

  for(let i = 0; i< numEnemies; i++) {

    // enemy movement
    enemies[i].y += enemies[i].speed;

    // check we haven't passed min or max Y
    let conditionUp = enemies[i].speed < 0 && enemies[i].y <= this.enemyMinY;
    let conditionDown = enemies[i].speed > 0 && enemies[i].y >= this.enemyMaxY;

    // if we passed the upper or lower limit, reverse
    if(conditionUp || conditionDown) {
      enemies[i].speed *= -1;
    }

    // check enemy overlap
    let enemyRect = enemies[i].getBounds();

    if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, enemyRect)) {
      console.log('Game over!');

      // end game
      return this.gameOver();
    }
  }
};

gameScene.gameOver = function() {
```

```
  // initiated game over sequence
  this.isTerminating = true;

  // shake camera
  this.cameras.main.shake(500);

  // listen for event completion
  this.cameras.main.on('camerashakecomplete', function(camera, effect){

    // fade out
    this.cameras.main.fade(500);
  }, this);

  this.cameras.main.on('camerafadeoutcomplete', function(camera, effect){
    // restart the Scene
    this.scene.restart();
  }, this);


};

// set the configuration of the game
let config = {
  type: Phaser.AUTO, // Phaser will use WebGL if available, if not it will use Canvas
  width: 640,
  height: 360,
  scene: gameScene
};

// create a new game, pass the configuration
let game = new Phaser.Game(config);
```

Learning Summary

- Camera effects
  - Access main camera: this.camera.main
  - Built-in events:fade, shake, flash
  - Listening for completion:

```
// shake camera
this.cameras.main.shake(500);

// listen for event completion
this.cameras.main.on('camerashakecomplete', function(camera, effect){

  // fade out
  this.cameras.main.fade(500);
```


In this lesson, you can find the full source code for the project used within the course. Feel free to use this lesson as needed – whether to help you debug your code, use it as a reference later, or expand the project to practice your skills!

You can also download all project files from the **Course Files** tab via the project files or downloadable PDF summary.

index.html

Found in Project/08 – Camera Effects

This HTML document is a game development project layout that integrates Phaser.js for game development. The Phaser framework and the main game code, both located in the 'js' directory, are linked in the body of the document. The canvas is set to span the full width of the page.

```
<!doctype html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <title>Learn Game Development at Zenva.com</title>
    <style>
      canvas {
        width: 100%;
      }
    </style>
</head>
<body>
  <script src="js/phaser.js"></script>
  <script src="js/game.js"></script>
</body>
</html>
```

game.js

Found in Project/08 – Camera Effects/js

This is a simple game programmed using Phaser framework where there's a player, enemies, and a goal. The game initialises parameters such as player/enemy speed and boundary limits, then preloads and renders images for the background, player, enemy, and goal. Gameplay involves the player moving towards the goal while avoiding enemies, with a game over condition when the player collides with an enemy or reaches the goal, after which the game resets.

```
// create a new scene
let gameScene = new Phaser.Scene('Game');

// initiate scene parameters
gameScene.init = function() {
  // player speed
  this.playerSpeed = 3;

  // enemy speed
  this.enemyMinSpeed = 2;
  this.enemyMaxSpeed = 4.5;
```

```
  // boundaries
  this.enemyMinY = 80;
  this.enemyMaxY = 280;

  // we are not terminating
  this.isTerminating = false;
};

// load assets
gameScene.preload = function(){
  // load images
  this.load.image('background', 'assets/background.png');
  this.load.image('player', 'assets/player.png');
  this.load.image('enemy', 'assets/dragon.png');
  this.load.image('goal', 'assets/treasure.png');
};

// called once after the preload ends
gameScene.create = function() {
  // create bg sprite
  let bg = this.add.sprite(0, 0, 'background');

  // change the origin to the top-left corner
  bg.setOrigin(0,0);

  // create the player
  this.player = this.add.sprite(40, this.sys.game.config.height / 2, 'player');

  // we are reducing the width and height by 50%
  this.player.setScale(0.5);

  // goal
  this.goal = this.add.sprite(this.sys.game.config.width - 80, this.sys.game.config.h
eight / 2, 'goal');
  this.goal.setScale(0.6);

  // enemy group
  this.enemies = this.add.group({
    key: 'enemy',
    repeat: 5,
    setXY: {
      x: 90,
      y: 100,
      stepX: 80,
      stepY: 20
    }
  });

  // setting scale to all group elements
  Phaser.Actions.ScaleXY(this.enemies.getChildren(), -0.4, -0.4);

  // set flipX, and speed
  Phaser.Actions.Call(this.enemies.getChildren(), function(enemy){
    // flip enemy
```



```
    enemy.flipX = true;

    // set speed
    let dir = Math.random() < 0.5 ? 1 : -1;
    let speed = this.enemyMinSpeed + Math.random() * (this.enemyMaxSpeed - this.enemy
MinSpeed);
    enemy.speed = dir * speed;

  }, this);
};

// this is called up to 60 times per second
gameScene.update = function(){

  // don't execute if we are terminating
  if(this.isTerminating) return;

  // check for active input (left click / touch)
  if(this.input.activePointer.isDown) {
    // player walks
    this.player.x += this.playerSpeed;
  }

  // treasure overlap check
  let playerRect = this.player.getBounds();
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  if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, treasureRect)) {
    console.log('reached goal!');

    // end game
    return this.gameOver();
  }

  // get enemies
  let enemies = this.enemies.getChildren();
  let numEnemies = enemies.length;

  for(let i = 0; i< numEnemies; i++) {

    // enemy movement
    enemies[i].y += enemies[i].speed;

    // check we haven't passed min or max Y
    let conditionUp = enemies[i].speed < 0 && enemies[i].y <= this.enemyMinY;
    let conditionDown = enemies[i].speed > 0 && enemies[i].y >= this.enemyMaxY;

    // if we passed the upper or lower limit, reverse
    if(conditionUp || conditionDown) {
      enemies[i].speed *= -1;
    }

    // check enemy overlap
    let enemyRect = enemies[i].getBounds();
```

```
    if(Phaser.Geom.Intersects.RectangleToRectangle(playerRect, enemyRect)) {
      console.log('Game over!');

      // end game
      return this.gameOver();
    }
  }
};

gameScene.gameOver = function() {

  // initiated game over sequence
  this.isTerminating = true;

  // shake camera
  this.cameras.main.shake(500);

  // listen for event completion
  this.cameras.main.on('camerashakecomplete', function(camera, effect){

    // fade out
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  this.cameras.main.on('camerafadeoutcomplete', function(camera, effect){
    // restart the Scene
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// set the configuration of the game
let config = {
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